NPTEL ONLINE CERTIFICATION COURSES

ARTIFICIAL INTELLIGENCE FOR ECONOMICS

Lecture – 11

Topic: Decision Tree and Random Forest



PROF. ADWAY MITRA

CENTRE OF EXCELLENCE IN ARTIFICIAL INTELLIGENCE

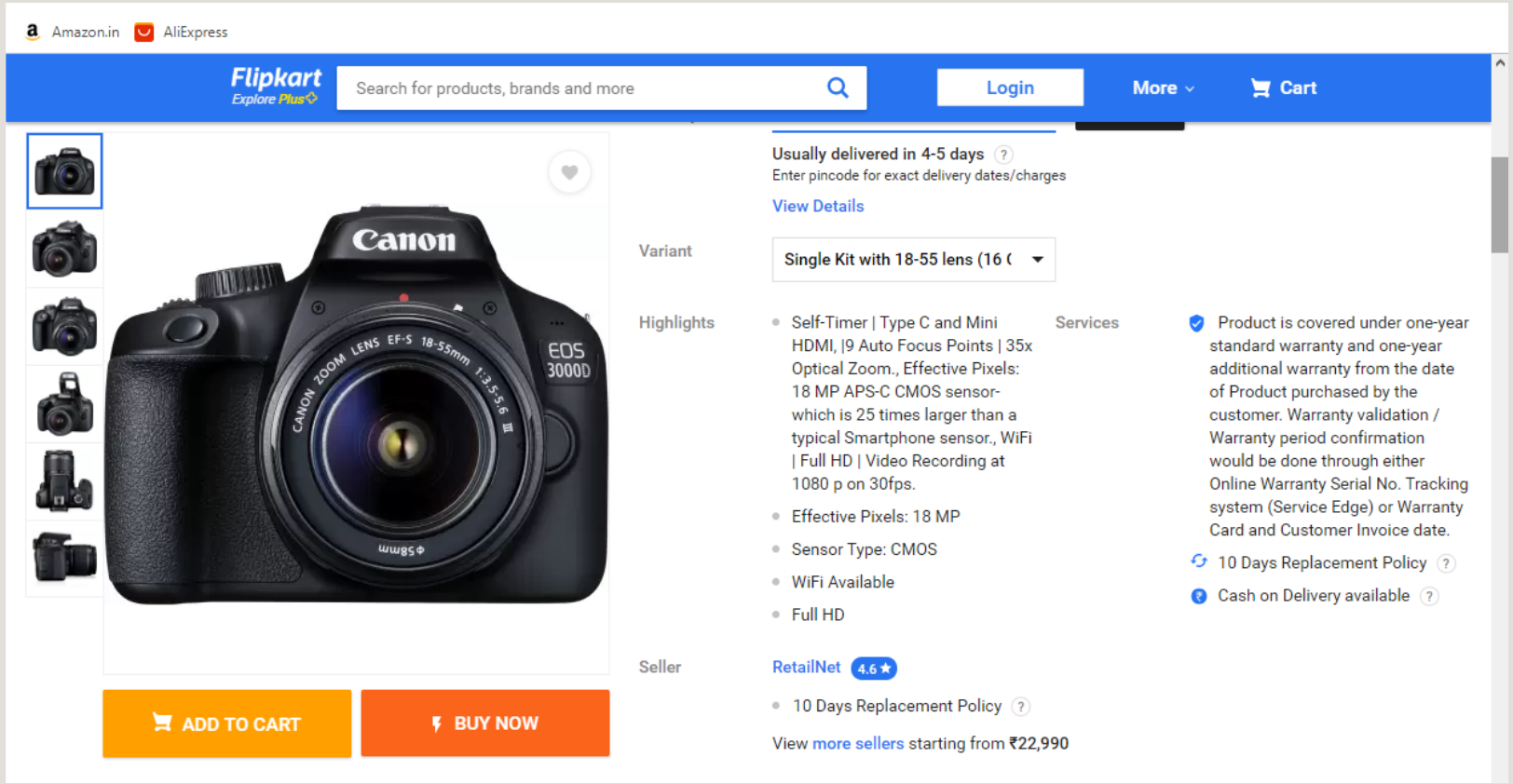
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CONCEPTS COVERED

- **Discriminative Features**
- **Entropy and Information Gain**
- **Classification and Regression Tree**
- **Random Forest**

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Product Ratings based on Features



- How much rating will a particular user give this camera out of 5?
- Probably depends on features!
- Which features does the user like?

Feature Selection

- The user has exactly 5 options: 1, 2, 3, 4 or 5 stars!
- Her choice depends on the different features of the product!
- But she may consider some features to be more important than others !
- Which features determine her vote?

Company	Color	Resolution	Video Rate	Price	Her Rating
C1	Black	10 MP	25 fps	$200	2
C1	White	15 MP	25 fps	$250	2
C2	White	12 MP	30 fps	$250	4
C1	Black	15 MP	30 fps	$300	3
C2	Black	20 MP	25 fps	$400	3
C2	White	12 MP	50 fps	$500	5
C2	Black	15 MP	30 fps	$250	????

Decision Tree for Feature Selection

Which features does she consider as important while rating?
Let's look at her history of rating 100 cameras!

Rating	Count
1	15
2	18
3	10
4	5
5	6

Company = C1,
Count=54

Rating	Count
1	6
2	6
3	8
4	15
5	11

Company = C2,
Count=46

Rating	Count
1	15
2	20
3	13
4	12
5	10

Color=Black,
Count=70

Rating	Count
1	6
2	4
3	5
4	8
5	7

Color=White,
Count=30

Which feature is more important for ratings - company or color???

What's a discriminative feature?

Prob(Y = HIGH | COMPANY = C1) = 11/54 ~ 0.2 [Easy to decide]
Prob(Y = HIGH | COMPANY = C2) = 26/46 ~ 0.55
COMPANY: good feature
Prob(Y = HIGH | PRICE < 300) = 25/70 ~ 0.36
Prob(Y = HIGH | PRICE >= 300) = 12/30 = 0.4
PRICE<300: bad feature
Prob(Y = HIGH | PRICE < 500) = 35/90 ~ 0.4
Prob(Y = HIGH | PRICE >= 500) = 2/10 ~ 0.2 [Easy to decide][Very few examples]
PRICE<500: doubtful feature

Decision Tree Algorithm

Idea: identify the “most discriminative” feature, use it to classify!

Problem 1: How to quantify “discriminative-ness”?

- entropy!

P(Y=1) = 0.5, p(Y=2) = 0.5 : low discriminative ability
P(Y=1) = 0.9, p(Y=2) = 0.1 : high discriminative ability

$$H = -\sum_i p_i (\log_2 p_i)$$

Case 1: H = 1 (0.69)
Case 2: H = 0.5 (0.33)

Problem 2: What if no feature is very discriminative?

- try a sequence of features!

Feature selection based on entropy

	COMPANY=C1	COMPANY=C2	NO SPLIT
#(Y=LOW)	43	20	63
#(Y=HIGH)	11	26	37
Entropy	0.51	0.68	0.66

Information gain =

Original Entropy – (Split1_size*Split1_ entropy + Split2_size*Split2_ entropy)

0.66 – (54/100*0.51 + 46/100*0.68) ~ 0.07

Feature selection based on entropy

	PRICE<300	PRICE>=300	NO SPLIT
#(Y=LOW)	45	18	63
#(Y=HIGH)	25	12	37
Entropy	0.65	0.67	0.66

Information gain =

Original Entropy – (Split1_size*Split1_ entropy + Split2_size*Split2_ entropy)

0.66 – (70/100*0.65 + 30/100*0.67) ~ 0!!

Feature selection based on entropy

	PRICE<500	PRICE>=500	NO SPLIT
#(Y=LOW)	55	8	63
#(Y=HIGH)	35	2	37
Entropy	0.67	0.5	0.66

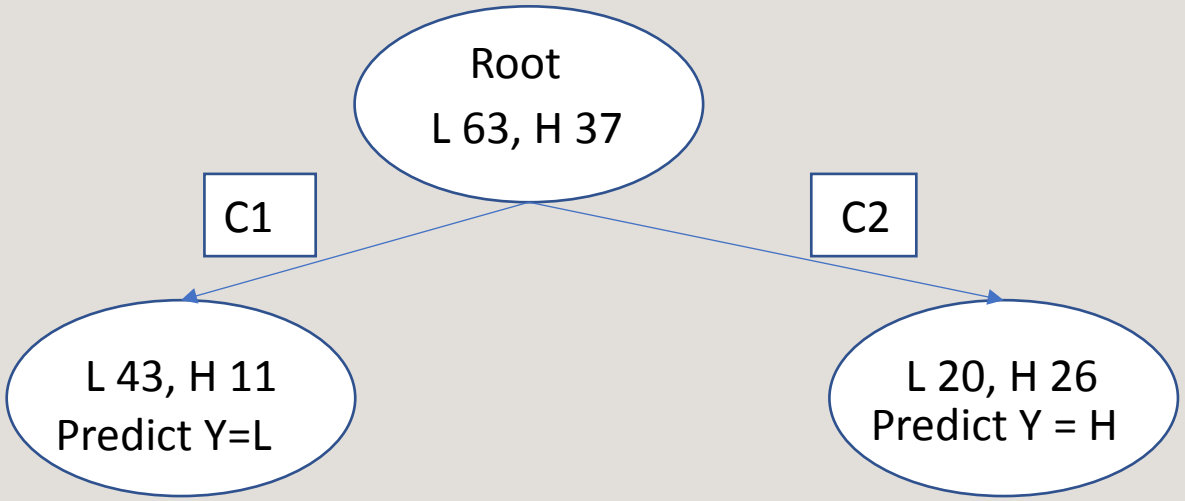
Information gain =

Original Entropy – (Split1_size*Split1_ entropy + Split2_size*Split2_ entropy)

0.66 – (90/100*0.67 + 10/100*0.5) ~ 0.01!!

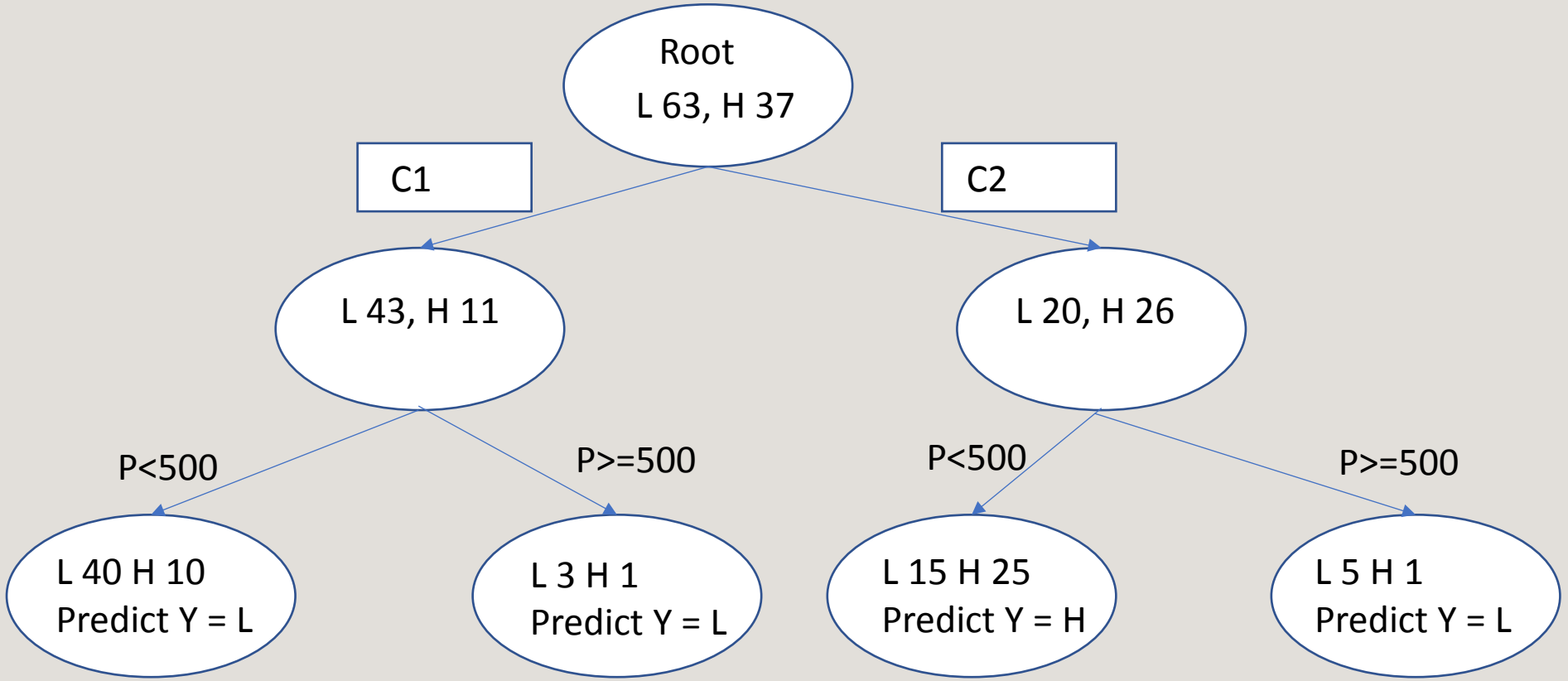
Feature selection based on entropy

Each discrete feature splits the dataset
Continuous features can always be converted to discrete
“Pure” dataset: - disbalanced class distribution
- low entropy
- high information gain
Choose that feature which provides most information gain!



Training accuracy: 43/63 for LOW, 26/37 for HIGH, 69/100 OVERALL

Decision Tree



Does this split provide “information gain”???

If yes, split. If no, stop at previous step

Decision Tree Algorithm

Train:

1. Identify the feature that results in maximum information gain
2. Split the dataset accordingly
3. Identify if any feature can result in further information gain on the split sets
4. If yes, split further. If no, stop.
5. Goto 3
6. At each leaf, the prediction is the mode label

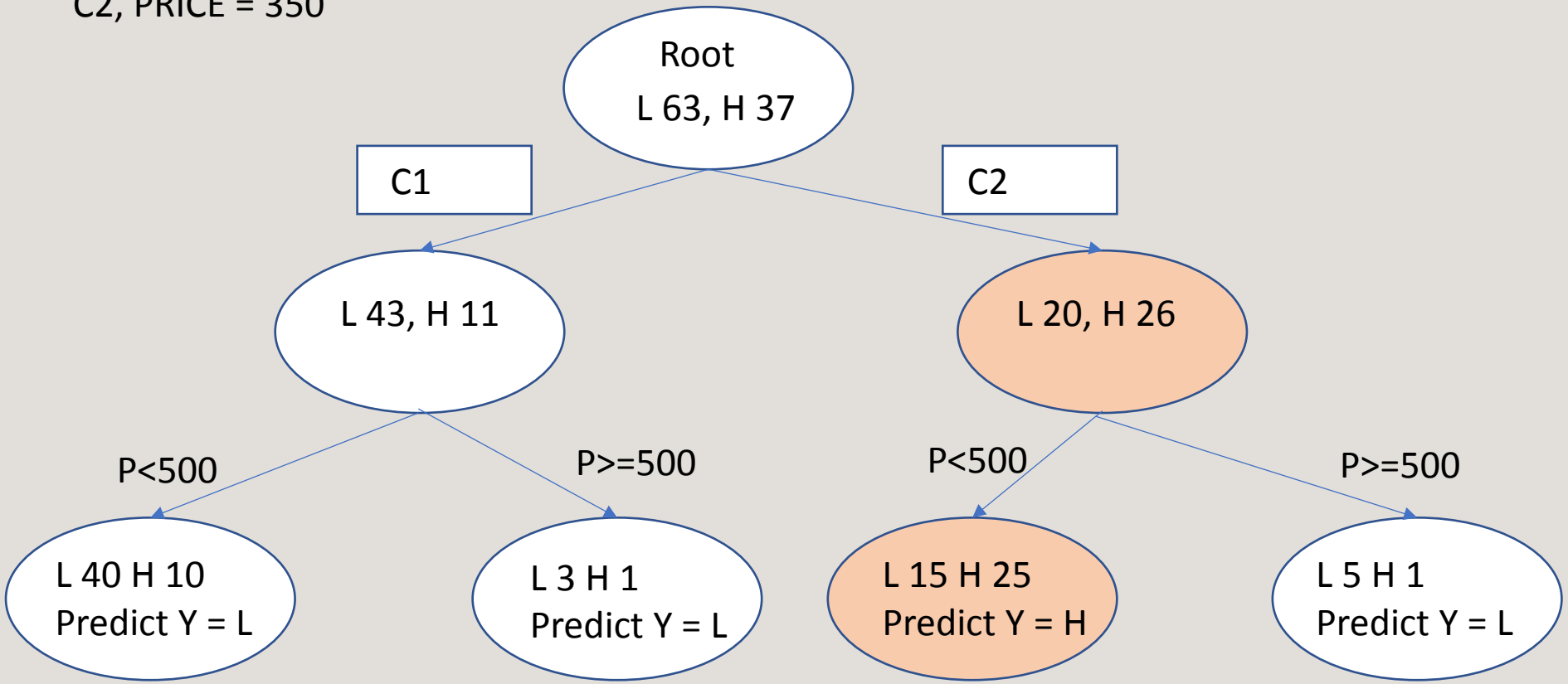
Test:

Follow the sequence of decisions based on the features of test example

Make prediction according to leaf

Decision Tree

TEST CASE: COMPANY =
C2, PRICE = 350



Prediction: Y= H

Decision Tree Algorithm

Advantage:

Easy to interpret

Easy to classify at test time

Provides a ranking of features (according to usefulness)

Disadvantages:

No optimal solution known, IG is just heuristic, can create many small branches

Can cause overfitting if tree grows deep (need to stop growing)

Decision Tree for Regression

What if we want to predict Average Rating of a product?

Real number between 1 and 5!

Decision trees can also be used for regression

Measure of homogeneity at each node: variance of labels (instead of entropy)

Split criteria: reduction in total variance (instead of information gain)

Final prediction: Mean label in the leaf node (instead of mode)

Ensembles of Trees

Ensemble Learning: Build a set of classifiers based on slightly different training criteria
For eg. different subsets of training examples
Another way: We can choose subsets of the features!
Each “version” has all the training examples, but only some of the features!

X(1)
X(2)
X(3)
X(4)
X(5)

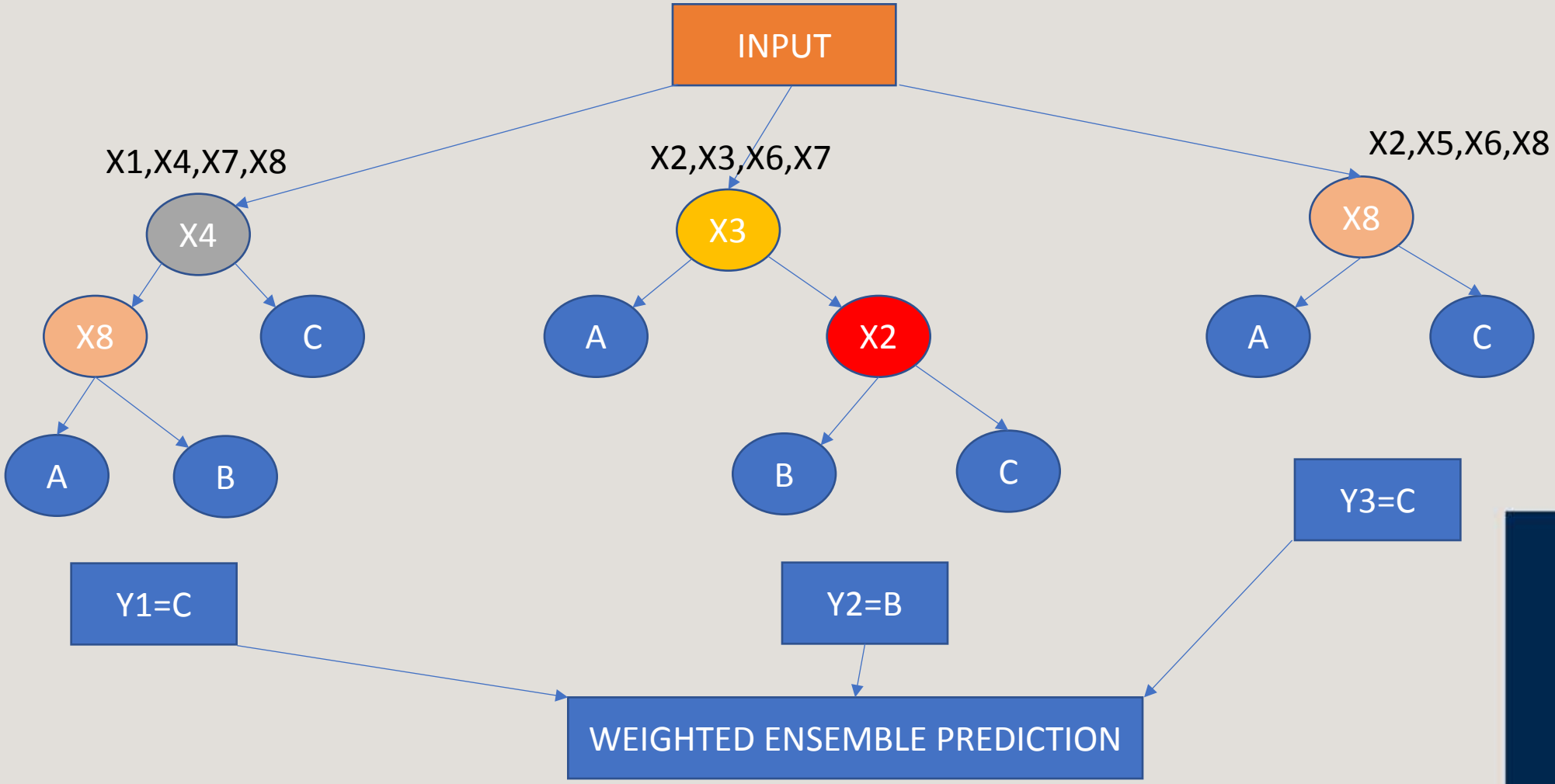
X(4)
X(2)
X(3)
X(3)
X(2)

X(2)
X(1)
X(4)
X(3)
X(3)

X(4)
X(2)
X(3)
X(2)
X(3)

Final classification: take “votes” from each classifier!
If the classifier used is Decision Tree, then this approach gives us a collection of trees!
This is called Random Forest - one of the more powerful classifiers around

Random Forest Classification



Possible Applications in Economics

Decision Trees are very preferred in many Economics applications due to their interpretable nature

Consumer Behavior Analysis: Predict customer-specific or overall popularity of an product

Market Basket Analysis: Analyze how sets of items may be merged to increase sales and consumer ease

Loan Default Prediction: Identify which customers are more/less likely to repay loans

Fraud Transaction Detection: Identify which credit card transaction sequences are likely to be fraudulant

Public Policy Impact Assessment: Predict possible impacts of a given sequence or collection of policy decisions on the economy of a country

CONCLUSION

- Some features are more useful than the others for predicting target values
- Entropy and Information Gain are used to evaluate the discriminative power of features
- Decision Tree specifies a sequence of features to be examined, based on which classification can be done
- Decision Tree can be extended for regression by using variance to identify feature discriminativeness
- Random Forest: Collection of Decision Trees trained on random subsets of features


THANK YOU

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ARTIFICIAL INTELLIGENCE FOR ECONOMICS

Lecture – 12

Topic: Linear Regression and Classifiers



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CONCEPTS COVERED

- **Supervised Learning – training and testing, loss function**
- **Linear Regression**
- **Linear Separability and Classification**
- **Algorithms for Linear Classification**

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Supervised Learning

- We are provided with observations, along with their “labels”: {y1, y2, ……. yN}
- Each observation is represented by a “feature vector”: {x1, x2, ……. xN}
- Feature vector belongs to “feature space”, label vector belongs to “label space”

- Aim: given a feature vector, predict its label!
- Need a function that maps from feature space to label space, i.e. y = f(x)!
- ‘f’ can be an explicit mathematical function, or some algorithm!

- Classification: label space discrete. Regression: label space continuous (real)

Approach for Supervised Learning

- We first make a “hypothesis” about function ‘f’ , e.g. $f(x) = x^Tx + a^Tx + b$
- We need to calibrate ‘f’ with the observations by choosing suitable parameters!
- e.g. finding suitable values of ‘a’, ‘b’ for function ‘f’

- We are provided with “training data” for calibration purposes
- Training data: {(x,y)} i.e. both feature vectors and true labels are known
- We choose a “loss function” that measures discrepancy between true and predicted values, l(y, f(x))
- Parameters chosen to minimize the total loss over training samples, $\Sigma l_i(y_i, f(x_i))$

- Validation: Calculate the loss on a different set of examples (not used for training)

Supervised Learning



FEATURE 1 = X1
FEATURE 2 = X2
FEATURE 3 = X3
FEATURE 4 = X4
....
....
FEATURE D = XD



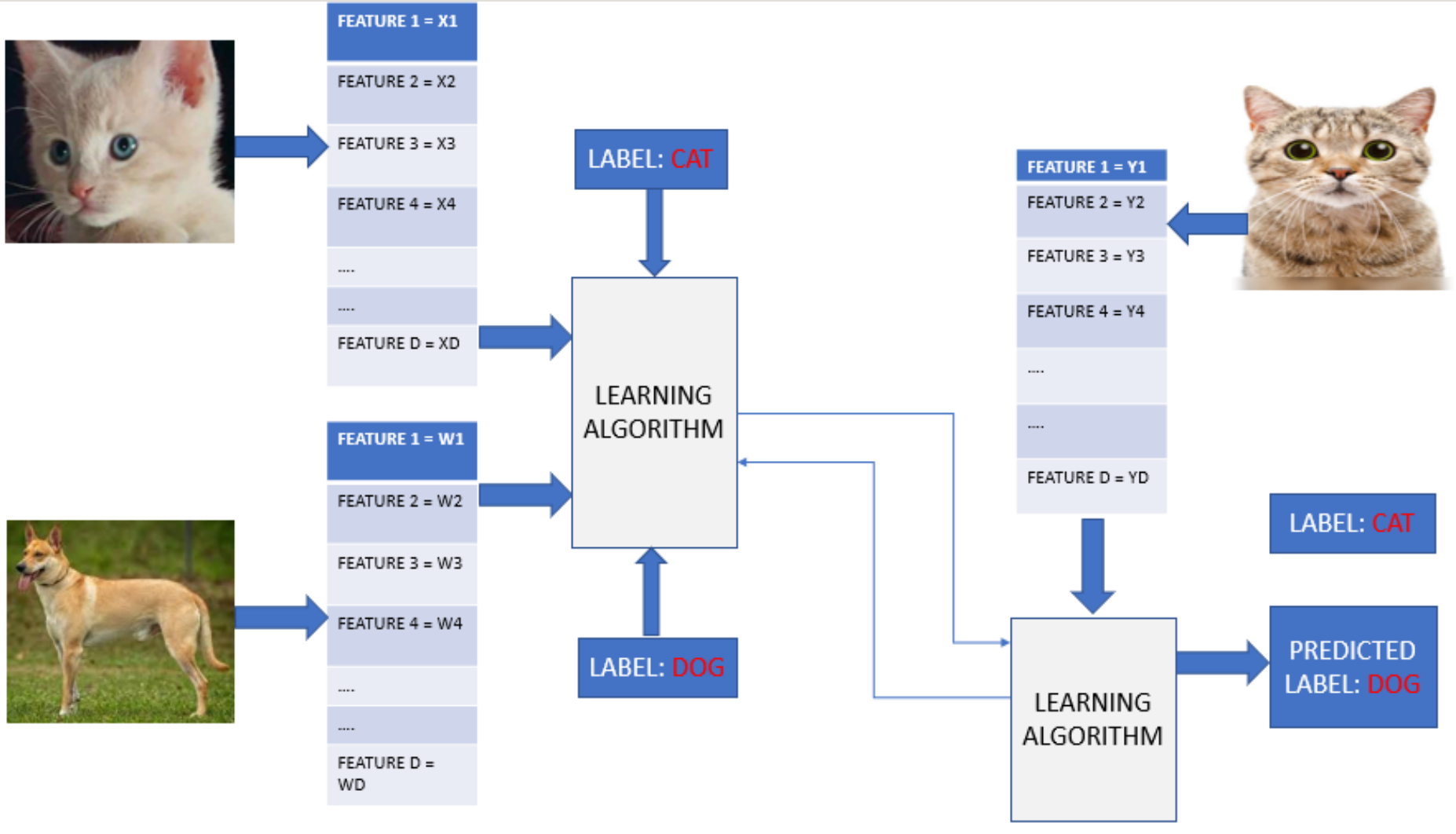
FEATURE 1 = W1
FEATURE 2 = W2
FEATURE 3 = W3
FEATURE 4 = W4
....
....
FEATURE D = WD

LEARNING ALGORITHM

LABEL: CAT

LABEL: DOG

Supervised Learning



Supervised Learning



FEATURE 1 = X1
FEATURE 2 = X2
FEATURE 3 = X3
FEATURE 4 = X4
....
....
FEATURE D = XD

LABEL: CAT



FEATURE 1 = W1
FEATURE 2 = W2
FEATURE 3 = W3
FEATURE 4 = W4
....
....
FEATURE D = WD

LEARNING ALGORITHM

LABEL: DOG



FEATURE 1 = Y1
FEATURE 2 = Y2
FEATURE 3 = Y3
FEATURE 4 = Y4
....
....
FEATURE D = YD

LEARNING ALGORITHM

LABEL: CAT

PREDICTED LABEL: CAT

Linear Regression

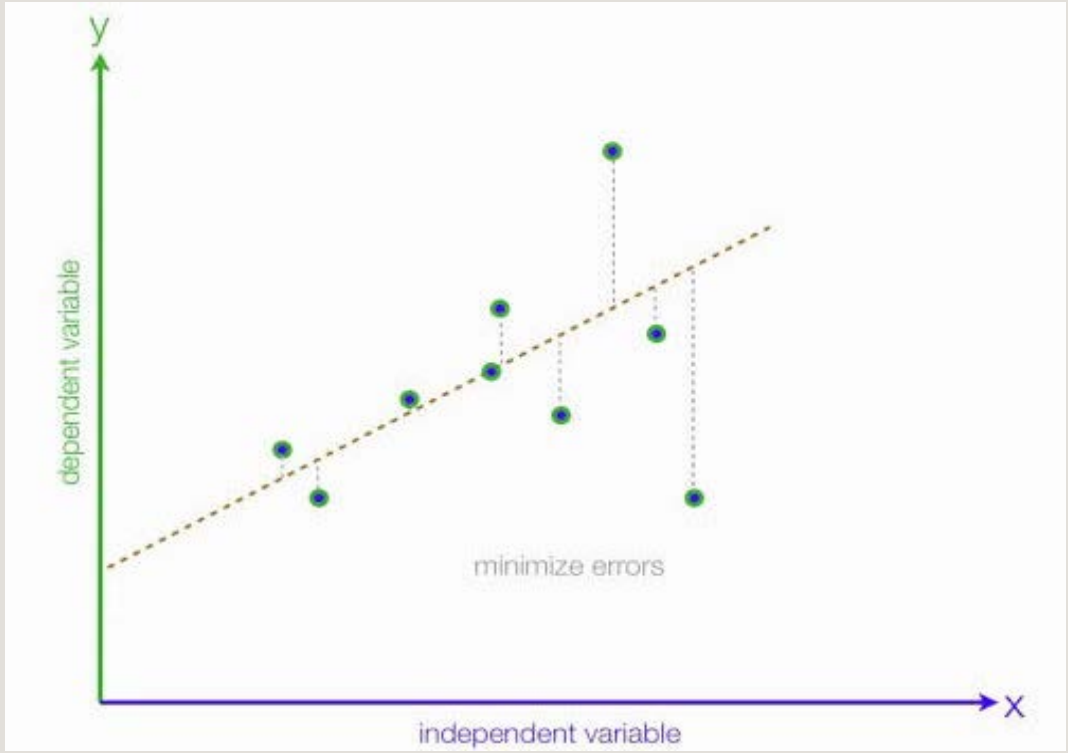
Suppose feature space is the D-dimensional real space.
Simplest assumption on f: a linear function!
Model: $y = f(x) = w^T x + w_0$

Bias term w_0 can be "absorbed" into weight vector: $f(x) = w^T x$
Task: estimate the weight vector!

1D features: Equivalent to fitting all observations on a single line

Higher dimension: fitting them onto a d-dimensional plane (hyperplane)

Squared Error Loss function:
$L(yi, hi) = (y_i - h_i)^2 = \sum_j (y_i - w_j x_{ij})^2$



How do customers rate a product?

User 1:

Title:

Presenter:

Date: Time:

Your job classification: ☐ Classified ☐ Professional/Technical ☐ Administrator ☐ Faculty

Please circle the appropriate response for each statement:

	Excellent	Good	Fair	Poor	
1. The relevance of this topic to me was	4	3	2	1	3
2. The usefulness of materials was	4	3	2	1	3
3. The effectiveness of the presenter was	4	3	2	1	3
4. I expect the future usefulness of this topic to be	4	3	2	1	3
5. My overall evaluation of this session is	4	3	2	1	4

User 2:

Title:

Presenter:

Date: Time:

Your job classification: ☐ Classified ☐ Professional/Technical ☐ Administrator ☐ Faculty

Please circle the appropriate response for each statement:

	Excellent	Good	Fair	Poor	
1. The relevance of this topic to me was	4	3	2	1	3
2. The usefulness of materials was	4	3	2	1	5
3. The effectiveness of the presenter was	4	3	2	1	1
4. I expect the future usefulness of this topic to be	4	3	2	1	4
5. My overall evaluation of this session is	4	3	2	1	4

Linear Regression for Rating Prediction

- Each product has D features (f1, f2, …. , fD)
- The rating "yi" given by any user "i" may be a weighted average of her scores on the individual features (x_{i1}, x_{i2}, … , x_{iD})
- The weights (w_{i1}, w_{i2}, … , w_{iD}) may vary from one user to another according to their respective priorities

- Simplest model for user rating: $y_i = \sum_j w_{ij}x_{ij} + w_{0i}$ (w_{0i} : bias)

- Need to estimate the weights "w": N users x D features
- Too many parameters!!

- New approximate model: $y_i = \sum_j w_j x_{ij} + w_{0i}$, i.e. all users have equal weights!
- Bias term can be absorbed into weight vector

Linear Regression for Rating Prediction

- The model in this case: $h_i = \sum_j w_j x_{ij} = w^T x_i$ (hi: predicted rating)
- Measurement of fit: squared error loss function
- Loss for user i: $L(yi, hi) = (y_i - h_i)^2 = \sum_i (y_i - \sum_j w_j x_{ij})^2 = \sum_i (y_i - w^T x_i)^2$

- Choose w to minimize total loss $\sum_i L(yi, hi)$ over all N users!
- Differentiate the total loss w.r.t. w, equate to 0, and solve!

- It is easy to show that $w = (X^T X)^{-1}(X^T Y)$ where $X = [x_1\ x_2\ \ldots\ldots\ x_N]$, $Y = [y_1\ y_2\ \ldots\ldots\ y_N]$

Linear Regression for Rating Prediction

- Linear regression model: $y_i = \sum_j w_j x_{ij}$ i.e. all feature ratings contribute to the final rating
- But in the examples, only a small number of features seem to influence the final rating, other features have little importance
- In case 1: One element in “w” will have high value, other elements will have small values
- In case 2: All elements except one in “w” have 0 value, i.e. “w” is sparse!

- Feature selection: the task of identifying the “important” features
- Important feature: those which strongly influence the final ratings

- Can we convert these demands into mathematical formulations?

Linear Regression: Regularization

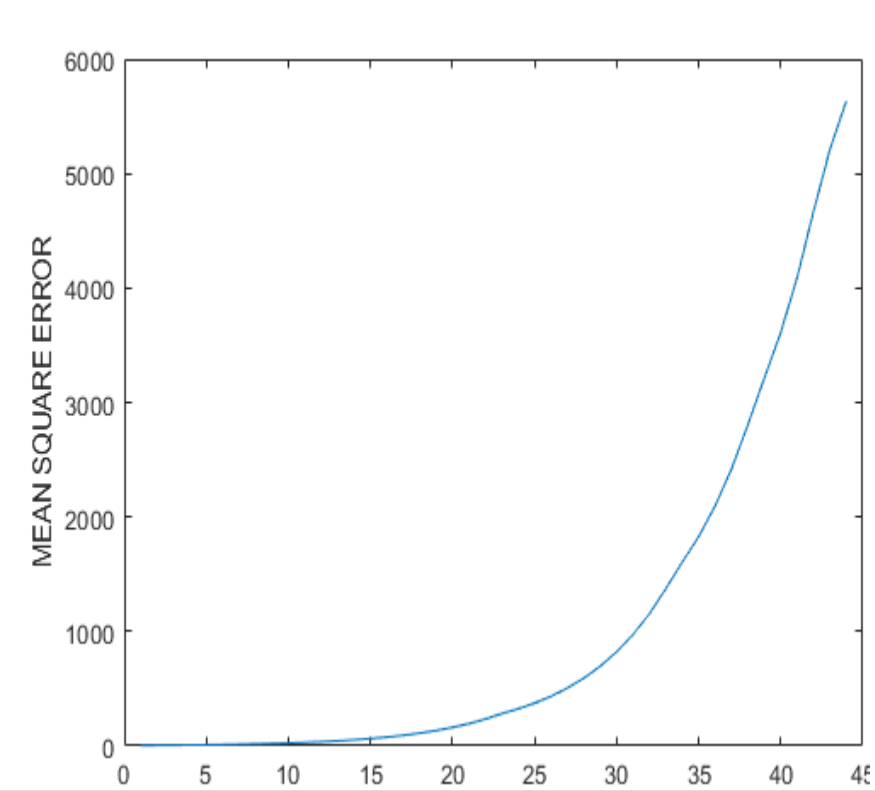
- General recipe: find a regularization function h(w)
- h(w) should have low value for suitable “w”, high for unsuitable “w”

- Find ‘w’ to minimize L(w) + λh(w)
- First term to find w that fits data, second term to find “w” that is suitable, λ to balance them!

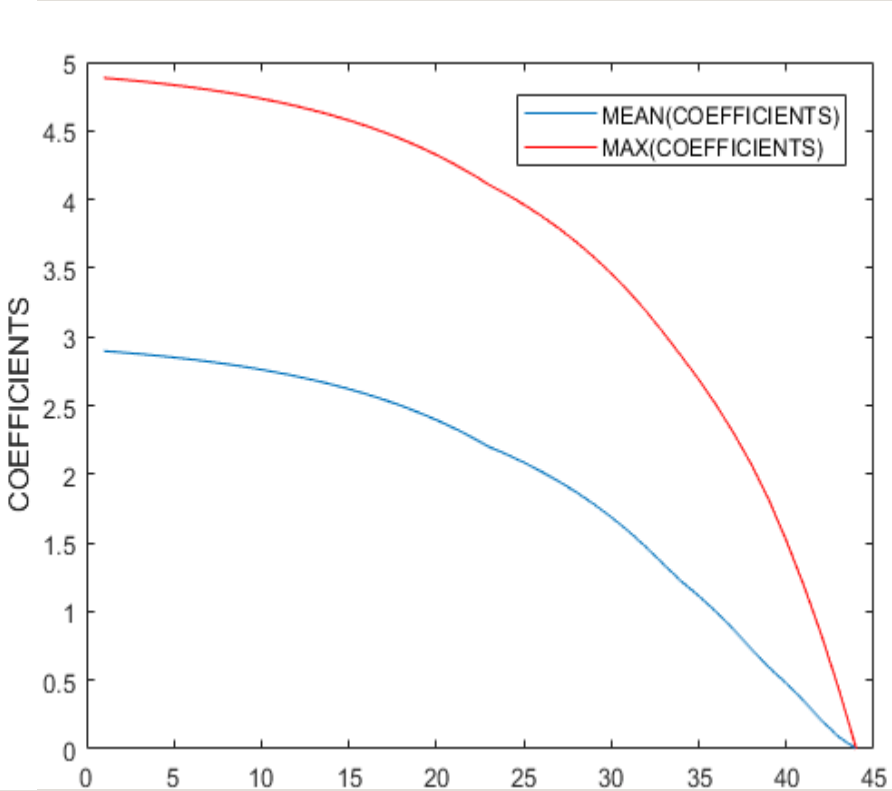
- Regularizer h(w) = $||w||_0$ promotes sparse “w”!
- However this regularizer is non-continuous and hence hard to optimization
- Relaxation: h(w) = $||w||_1$: can provide “almost sparse” solutions!

- Known as LASSO regression (Least Absolute Shrinkage and Selection Operator)
- Can be solved using numerical methods!

Linear Regression: Regularization



LAMBDA



LAMBDA

The parameter λ controls the relative importance of error minimization and structure imposition

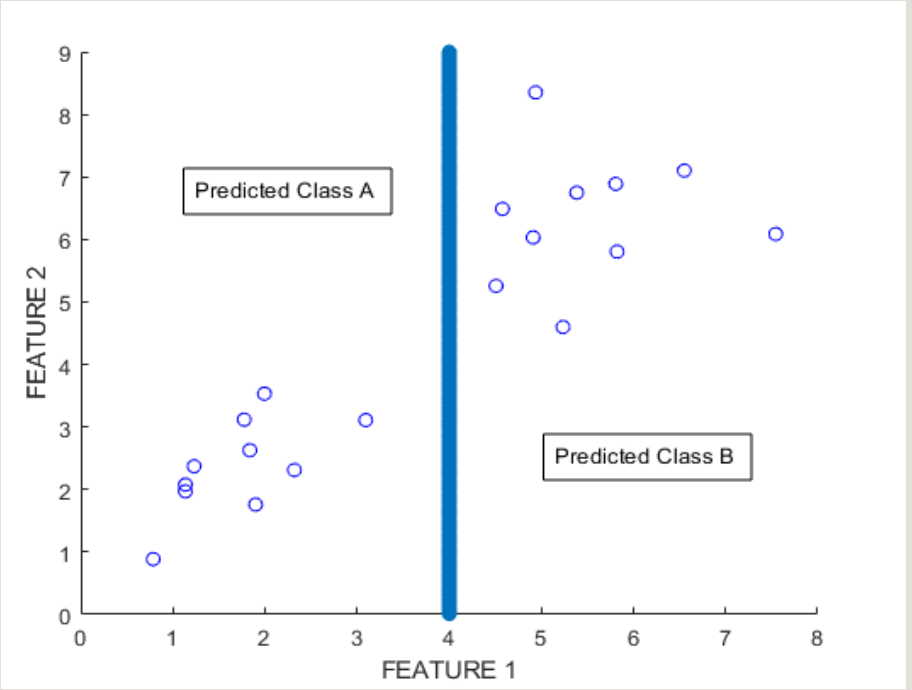
Linear Classification

- The same idea can be extended to classification: linear classifiers
- We consider binary classification: only two class labels available (+1, -1)

- Linear Classifier: a line/hyperplane in the feature space
- Divides the feature space into 2 parts (half-spaces)
- Data-points in each half-space considered to be of one class

- Model: $yi = \text{sign}(w^T x_i)$

- Task: estimate ‘w’
- Loss function: 0-1 (number of times predicted and true lables don’t match)!



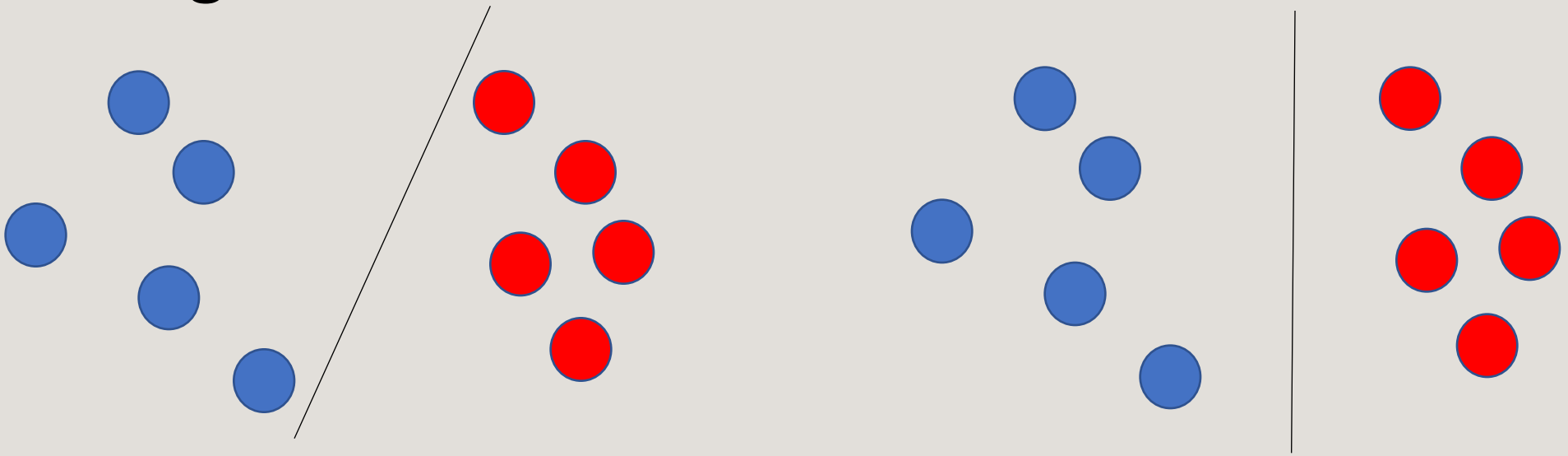
Finding a Linear Classification

- If there exists a linear classifier that achieves 100% accuracy, the dataset is called "linearly separable"
- Otherwise, find that linear classifier which achieves maximum accuracy!

- How to estimate 'w' from the training samples?
- Perceptron: a simple iterative algorithm
  - Starts with initial 'w'
  - Tries to classify all examples one-by-one
  - Corrects estimate upon each misclassification!

- Finds a 'w' if one exists
- But is the best 'w'?

Finding a Linear Classification



SOLUTION 1 **SOLUTION 2**

- In Solution 1, a small amount of feature noise can result in misclassification

- We prefer the linear classifier to be as far away from examples as possible (Solution 2)
- Max-Margin Linear Classifier!

- Can be estimated using Support Vector Machine algorithm
- Requires complex optimization techniques (quadratic program)

Application in Economics

- Both classification and regression are useful in various tasks of Economics
- Financial Risk Assessment: it is necessary to **classify** financial instruments or borrowers based on their risk profile. Such classification may be using linear classifier if the attributes are all real-valued
- Economic Forecasting: Many tasks in economics involve predict future economic variables, including stock market trends or GDP of a country. The feature vector can be constructed using various real-valued economic indicators. Such predictions can be either classification or regression tasks (depending on nature of target)
- Variable Selection: For any forecasting task, we may use large number of potential explanatory variables. LASSO regression helps in selecting the most relevant variables while excluding irrelevant ones. This not only helps in building more concise models, but also develop better understanding of economic processes

CONCLUSION

- Supervised Learning: learn a function that maps a vector of features/predictors to a target variable
- Linear Regression/Classification: the function is assumed to be linear
- Weight vector can be estimated by minimizing a suitable loss function
- Specific structure (like sparsity) can be imposed using regularization
- Linear classifier can be estimated using perceptron or Support Vector Machine
- Support Vector Machine gives us max-margin linear classifier


THANK YOU

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ARTIFICIAL INTELLIGENCE FOR ECONOMICS

Lecture – 13

Topic: Uncertainty Modeling



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CONCEPTS COVERED

- **Random Variables, distributions**
- **Expectation and Risk Analysis**
- **Bayes Theorem and Bayesian Networks**

Do not write here (Your Video will appear here)

Uncertainty Modeling in Economics

- Many economics decisions are based on estimates and projections for future
  - should I invest in a particular company? Will it do well?
  - If yes, how much should I invest?
  - How much budget should be allotted for a purpose? How may the expenditure change?

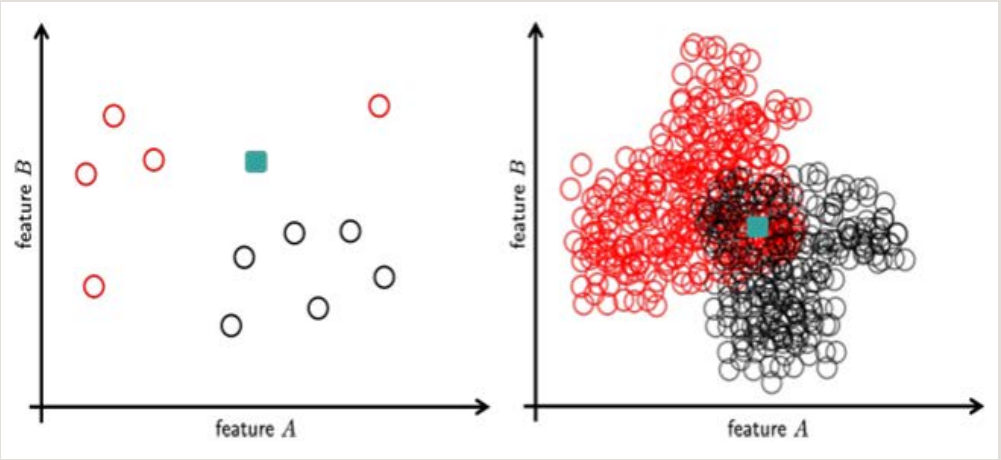
- Probability theory: attach a value to each possible outcome!

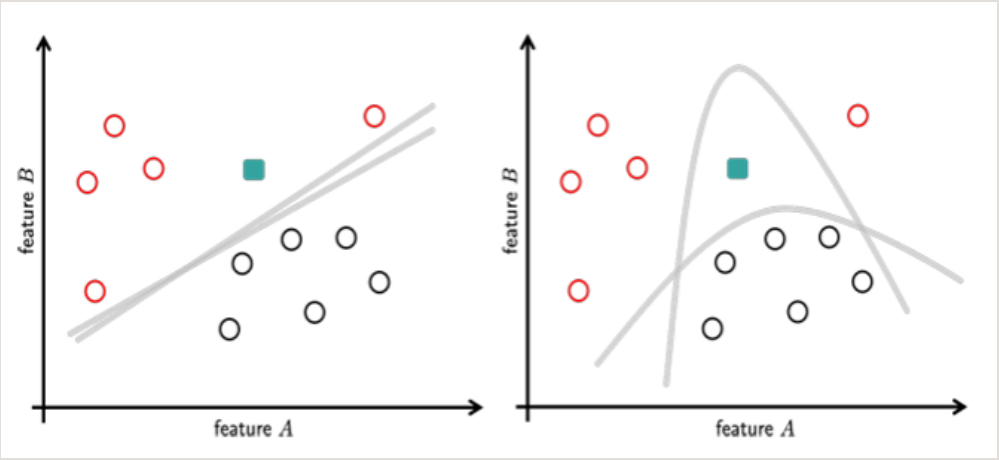
- Allows us to quantify
  - chances of success/failure
  - likely outcomes with a range of unlikely outcomes

Sources of Uncertainty

- Two main sources of uncertainty in most modelling and prediction tasks:
  Aleatoric and Epistemic Uncertainty

- Aleatoric Uncertainty: Arising due to imperfect or incomplete predictors
  - some observed features/predictors in X may be wrong due to measurement faults
  - some important predictors may not be observed or even observable!





Credits: Eyke Hüllermeier

- Epistemic: Arising due to unsuitable predictive model (lack of knowledge)
  - We may have assumed a linear model ‘f’ as y= f(x)
  - Maybe a quadratic model would have been more accurate!

Random Variables and Distributions

- Random Experiment: a process that can have several possible outcomes, but cannot say for certain exactly which outcome will happen
  - eg. a coin toss, a dice throw

- Sample Space □ : set of possible outcomes, Event: subsets of the sample space
  - eg. Coin: □ = {Head, Tail}, Dice: □ = {1, 2, 3, 4, 5, 6}
  - Dice event: odd/even, coin event: more than 5 heads in 10 tosses etc

- Probability measure: a mapping ‘p’ from event space to interval [0,1]
  Such that p(□) = 1, p(A) = 1 - p(A^c) where A^c = □ - A [A: any event]

- Random Variable X: a mapping from event space to real numbers

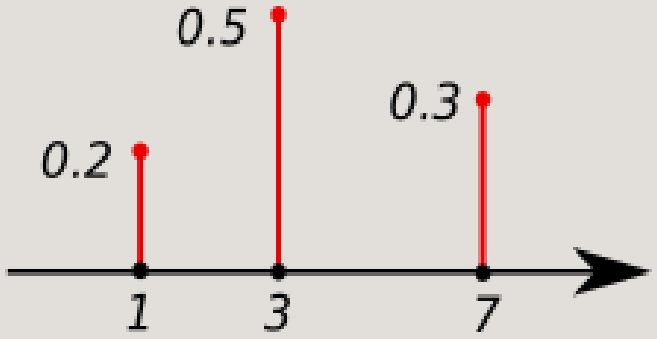
Mass and Density Functions

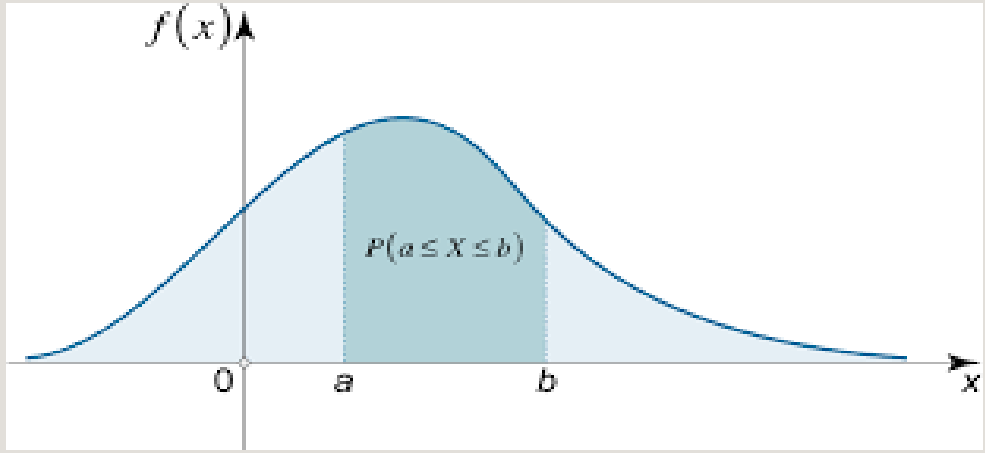
P(X<=a) = Σ p(w) where w is an event with X(w) <=a

Range of X (unique values): support of Random Variable X

Discrete Random Variable: support set is discrete
Continuous Random Variable: support set is real-valued

Set of p-values for different values of X: probability distribution of X!





Source: Wikipedia

Common Distributions

Cumulative Distribution Function (CDF): F(x) = p(X<x)

Each DRV is characterized by its Probability Mass Function (PMF): f(x) = p(X=x)

Distribution	Support	PMF	Parameters
Bernoulli	$\{0, 1\}$	$p^x(1-p)^{(1-x)}$	$0 \leq p \leq 1$
Binomial	$\mathcal{Z}$	$\binom{N}{x} p^x (1-p)^{N-x}$	$N \in \mathcal{Z}^+, 0 \leq p \leq 1$
Poisson	$\mathcal{Z}$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$\lambda \in \mathcal{R}^+$
Geometric	$\mathcal{Z}^+$	$(1-p)^{x-1}p$	$0 \leq p \leq 1$
Categorical	$\{V_1, \ldots, V_K\}$	$\prod_{k=1}^{K} p_k^{I(x=V_k)}$	$0 \leq (p_1, p_2, \ldots, p_K) \leq 1$ $\sum_{i=1}^{k} p_i = 1$
Multinomial	$\mathcal{Z}^K$	$\frac{N!}{x_1!..x_K!} \prod_{k=1}^{K} p_k^{x_k}$	$N \in \mathcal{Z}^+, \sum_{i=1}^{k} p_i = 1$ $0 \leq (p_1, \ldots, p_K) \leq 1$

Common Distributions

Cumulative Distribution Function (CDF): F(x) = p(X<x)

Each CRV is characterized by its Probability Density Function (PDF): f(x) = p(x-Δ<X<x+ Δ)

Distribution	Support	PDF	Parameters
Uniform	$\mathcal{R}$	$\frac{1}{b-a}$	$(a,b) \in \mathcal{R}$
Beta	$(0,1)$	$\frac{1}{B(a,b)} x^{(a-1)}(1-x)^{(b-1)}$	$(a,b) \in \mathcal{Z}^+$
Exponential	$\mathcal{R}^+$	$\lambda exp(-\lambda x)$	$\lambda \in \mathcal{R}+$
Gamma	$\mathcal{R}+$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} exp(-\beta x)$	$(\alpha,\beta) \in \mathcal{R}^+$
Gaussian	$\mathcal{R}$	$\frac{1}{\sqrt{(2\pi)}\sigma} exp(-\frac{1}{2\sigma^2}(x-\mu)^2)$	$\mu \in \mathcal{R}, \sigma \in \mathcal{R}^+$
χ^2	$\mathcal{R}^+$	$\frac{x^{K/2-1}e^{-x/2}}{2^{\frac{k}{2}}\Gamma(K/2)}$	$K \in \mathcal{Z}^+$

Conditional and Joint Distributions

- Joint probability: the probability that two or more events can occur simultaneously

- X, Y are any two random variables : their joint distribution
  $f_{XY}(x, y)= p(X=x, Y=y)$ if X, Y are discrete
  $f_{XY}(x, y)= p(x-\Delta<X< x+\Delta, y-\Delta<Y< y+\Delta)$ if X, Y are continuous

- Independent variables: $f_{XY}(x, y)= f_X(x)* f_Y(y)$

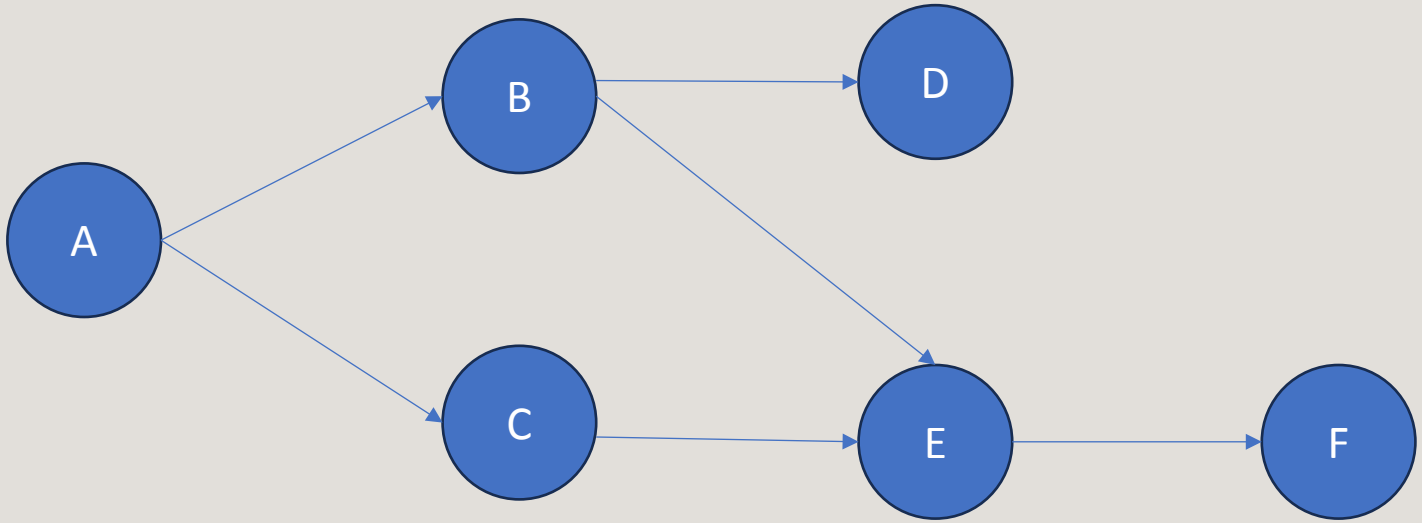
- Conditional probability: the probability that one event will happen, when we already know that another event has happened
  eg. probability that outcome of a dice throw is 2, when we already know that it is even

- $f_{Y|X}(y|x) = p(Y=y \mid X=x) = p(X=x, Y=y)/p(X=x)$ for discrete X, Y (similarly for continuous)

- Law of total probability: $f_X(x) = \Sigma_y f_{XY}(x, y)$ [discrete], $= \int_y y f_{XY}(x, y)$
- Bayesian Theorem: $f_X(x)* f_{Y|X}(y|x) = f_Y(y)* f_{X|Y}(x|y)$

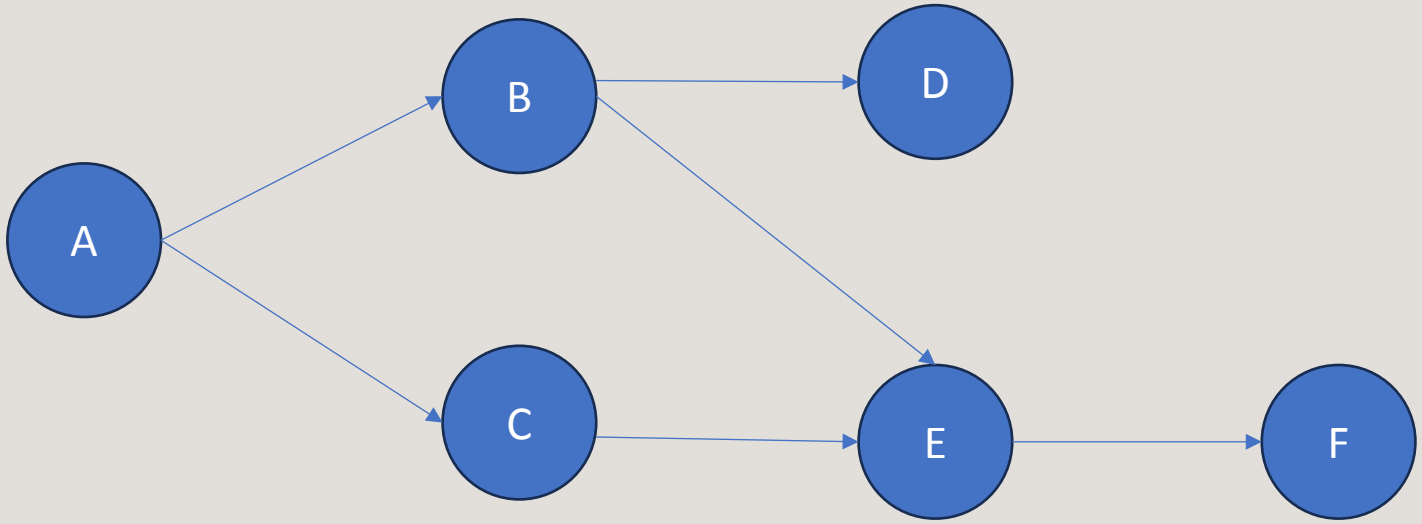
Bayesian Networks of Multiple Variables

- We can represent a complex system involving many uncertain but inter-dependent variables using these concepts
- Dependence relations between variables represented by conditional distributions
- Bayesian Network: a directed graph where each vertex represents a random variable



Bayesian Networks of Multiple Variables

- A conditional distribution defined at each node over its parent nodes
- Joint distribution of all variables: product of all these conditional distributions!



P(A=1) = 0.7, p(B=1|A=1) = 0.9, p(B=1|A=0) = 0.2, p(C=1|A=1) = 0.4, p(C=1|A=0) = 0.6,
P(E=1|B=1,C=1) = 0.5, P(E=1|B=1,C=0) = 0.25, P(E=1|B=0,C=1) = 0.1, P(E=1|B=0,C=0) = 0.9

Learning Tasks

- Probabilistic Inference: estimate possible values of any variable, conditioned on some others
- P(B=1|C=0, D=1) = ?

- Not specified by the model, but can be calculated using Bayes Theorem!

- Given independent observations of variable(s): X={x1, x2, …… xN}
- How do we fit a distribution to it/them?
  - Need to consider a known distribution family with support consistent with observations
  - Distribution's PDF should match data histogram
  - Calculate joint probability/density of all observations

    $p(X) = p(x1)*p(x2)*\ldots\ldots*p(xN)$
  - Maximize p(X) with respect to model parameters!

Applications in Economics

- **Risk Assessment and Management**: Bayesian networks can model complex dependencies between different economic variables and events, allowing for more accurate risk assessment and management. This is particularly useful in financial markets, where understanding the interplay between various factors such as interest rates, inflation, and geopolitical events is crucial for making informed decisions.

- **Market Analysis**: Bayesian networks can analyze market dynamics by modeling the relationships between different market variables such as supply, demand, prices, and consumer behavior.

- **Supply Chain Management**: Bayesian networks can model the dependencies between different stages of the supply chain, including production, distribution, and inventory management. This can help businesses optimize their supply chain operations by identifying potential bottlenecks, reducing risks, and improving efficiency.

Expectations

- Expected value of a random variable: it's probability-weighted average value!

- $E(X) = \Sigma x*f(x)$ (if X is a DRV)
  $= \int x*f(x)*dx$ (if X is a CRV)

- For eg. i) X ~ Bernoulli(p). Then $E(X) = 1*p + 0*(1-p) = p$
  ii) X ~ N(u, s). Then $E(X) = u$

- $E(X + Y) = E(X) + E(Y)$ where X, Y are any two random variable over same support
  $E(X*Y) = E(X)*E(Y)$ if X, Y are independent

- Variance $V(X) = E(X-E(X))^2 = E(X^2) – (E(X))^2$
  Provides a range of possible values of X, around the mean!

- For example, $V(X) = p(1-p)$ if X ~ Bernoulli (p)
  $= s$ if X ~ N(u, s)

Expectation-based Risk Analysis

- I have 100 rupees to bet on a cricket match. If I bet ‘x’ rupees on Team A, I win ‘2x’ if they win
- but I lose ‘20x’ rupees if they lose. Probability that ‘A’ will win: p.
- What is my expected gain if I bet ‘x’?
- How much should I bet so that I will be left with at least 10 rupees?
- How much should I bet so that I have a chance of winning 150 rupees?

- I want to predict the stock price Y of a company, based on some economic predictors X

- I have observations {(X, Y)} using which I can
  - i) train a Model: Y = f(X) by supervised learning (linear regression)
  - ii) Fit a model p(Y|X) using parameter estimation

- Suppose the economic indicators have value ‘x’.
  - i) What is the expected value of the stock price?: f(x) or E(Y|X=x)
  - ii) What is the risk that the stock price may fall below ‘y’? P(Y<y | X=x)

CONCLUSION

- Uncertainty quantification is very important in economic decision-making
- We represent various factors as random variables and consider them to follow a known family of distributions
- Family and parameter values chosen based on observations of each variable
- Relations between different variables can be represented through conditional distributions
- Expectation captures “average-case” scenario, variance suggests how much deviation from expectation is possible!


THANK YOU

NPTEL ONLINE CERTIFICATION COURSES

ARTIFICIAL INTELLIGENCE FOR ECONOMICS

Lecture – 14

Topic: Neural Networks



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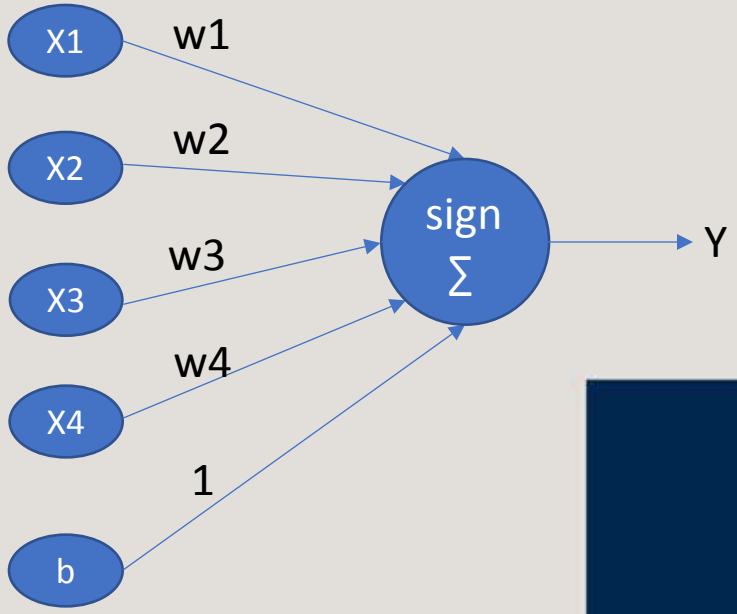
CONCEPTS COVERED

- **Non-linear Classification**
- **Neural Network structure and components**
- **Neural Network as function approximator**
- **Backpropagation**

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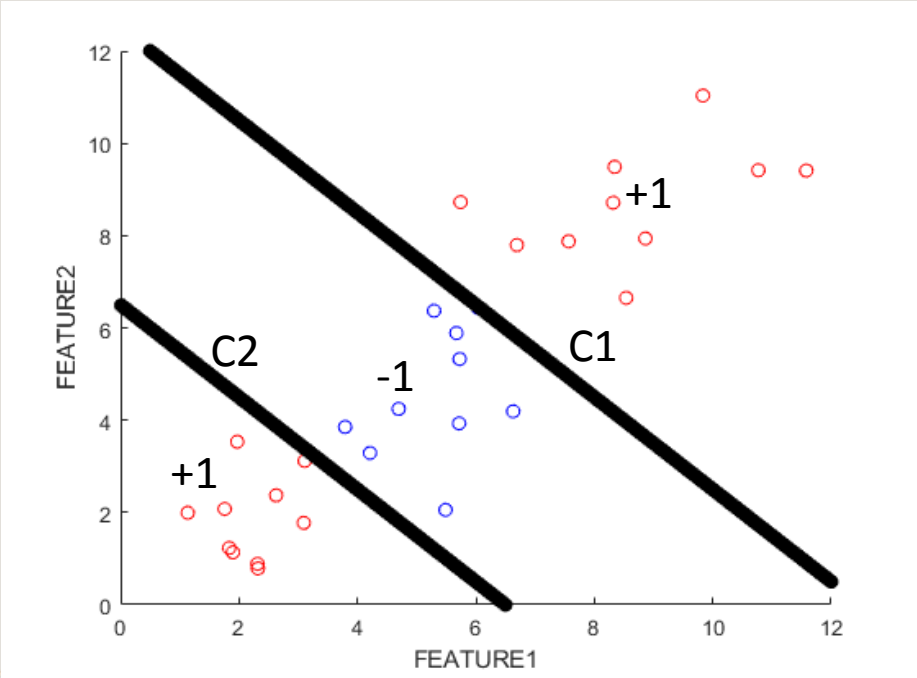
Neural Network as Linear Classifier

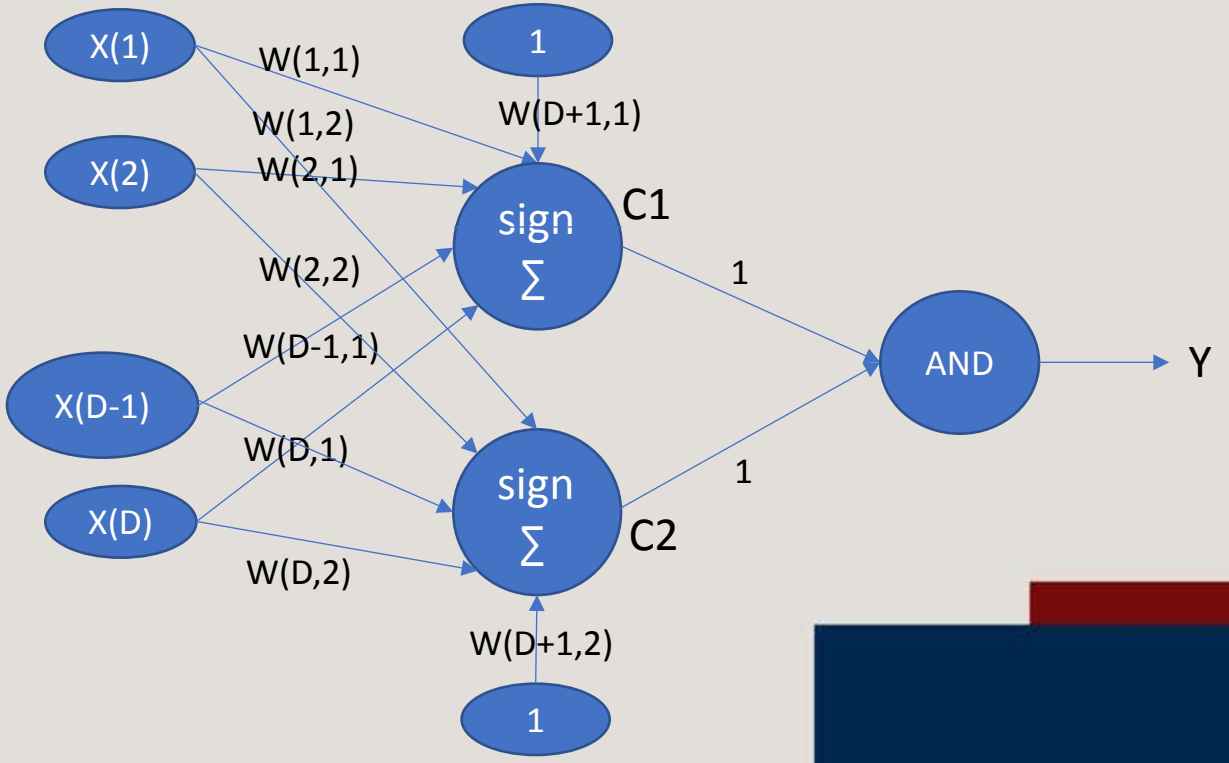
- A linear classifier computes a weighted sum of the features and a bias
- Then runs the “sign” function on the result
- Y = sign(w.x)
- We can represent this by a graph
- Each input dimension: one input node
- Each input node connected to output node
- Each connection edge carries a weight



Neural Network as Multi-Linear Classifier

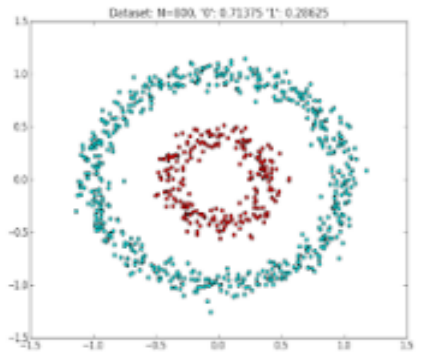
- A single Linear classifier is often not enough!
- We may need a combination of linear classifiers!
- If C1:+1 OR C2:+1, Final:+1
- If C1:-1 AND C2:-1, Final: -1





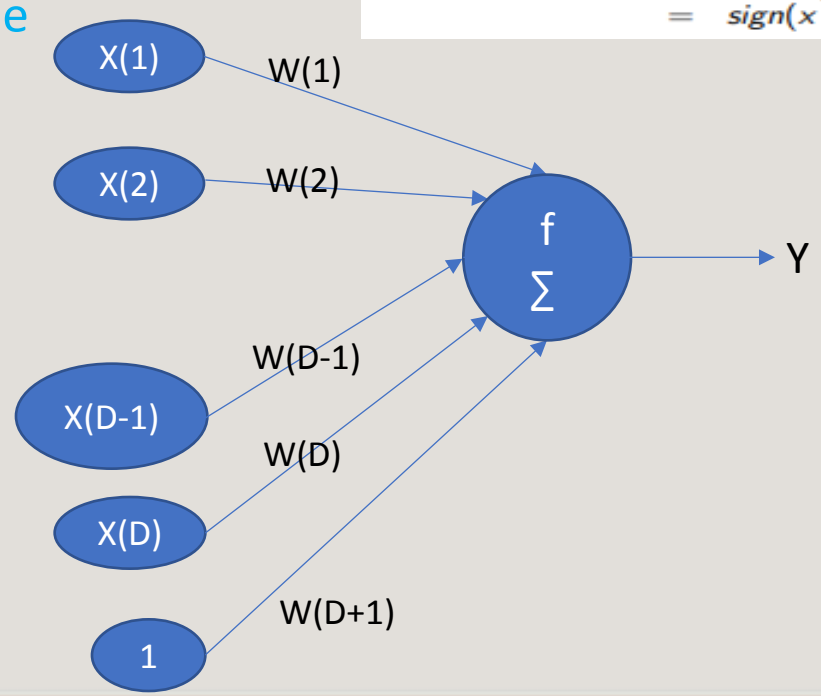
Neural Network as Non-Linear Classifier

- A linear classifier computes a weighted sum of the features and a bias
- Then runs the “sign” function on the result
- Y = f(W.X), where f is non-linear function
- We can represent this by a graph
- Each input dimension: one input node
- Each input node connected to output node
- Each connection edge carries a weight

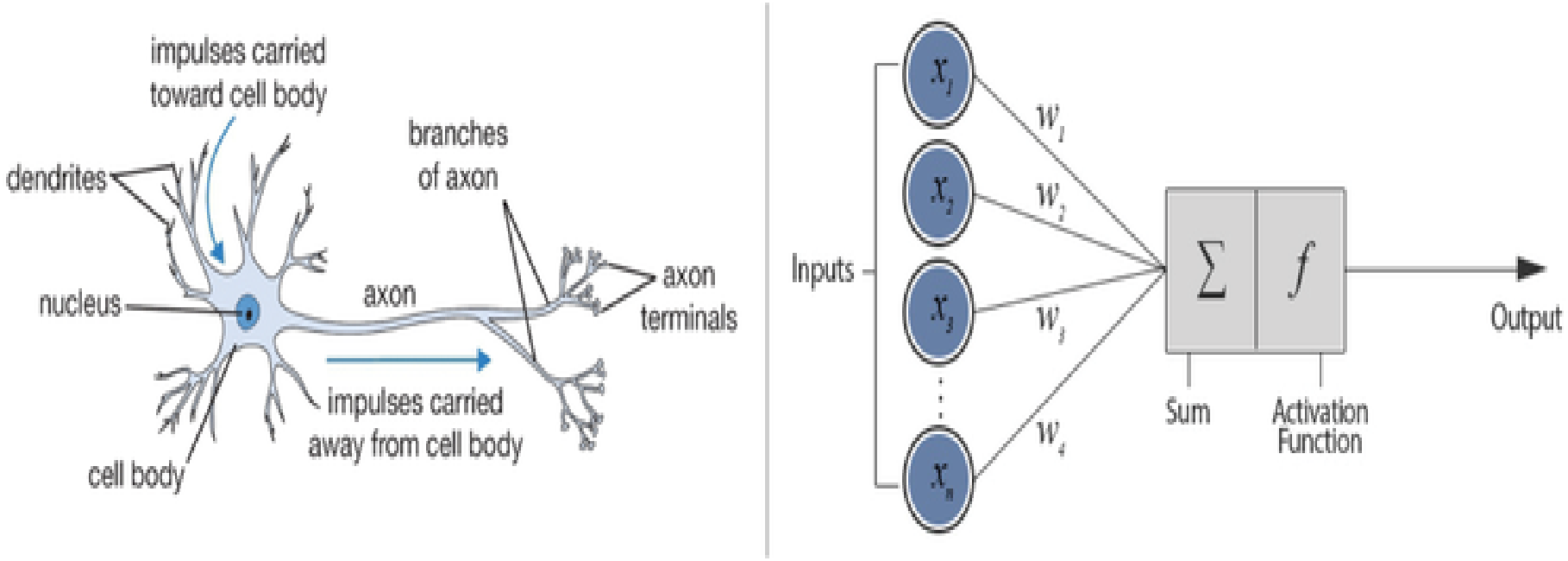


Separating structure:

$$y = sign((x - x_0)^T (x - x_0) - R)$$
$$= sign(x^T x - 2x_0^T x + x_0^T x_0 - R)$$

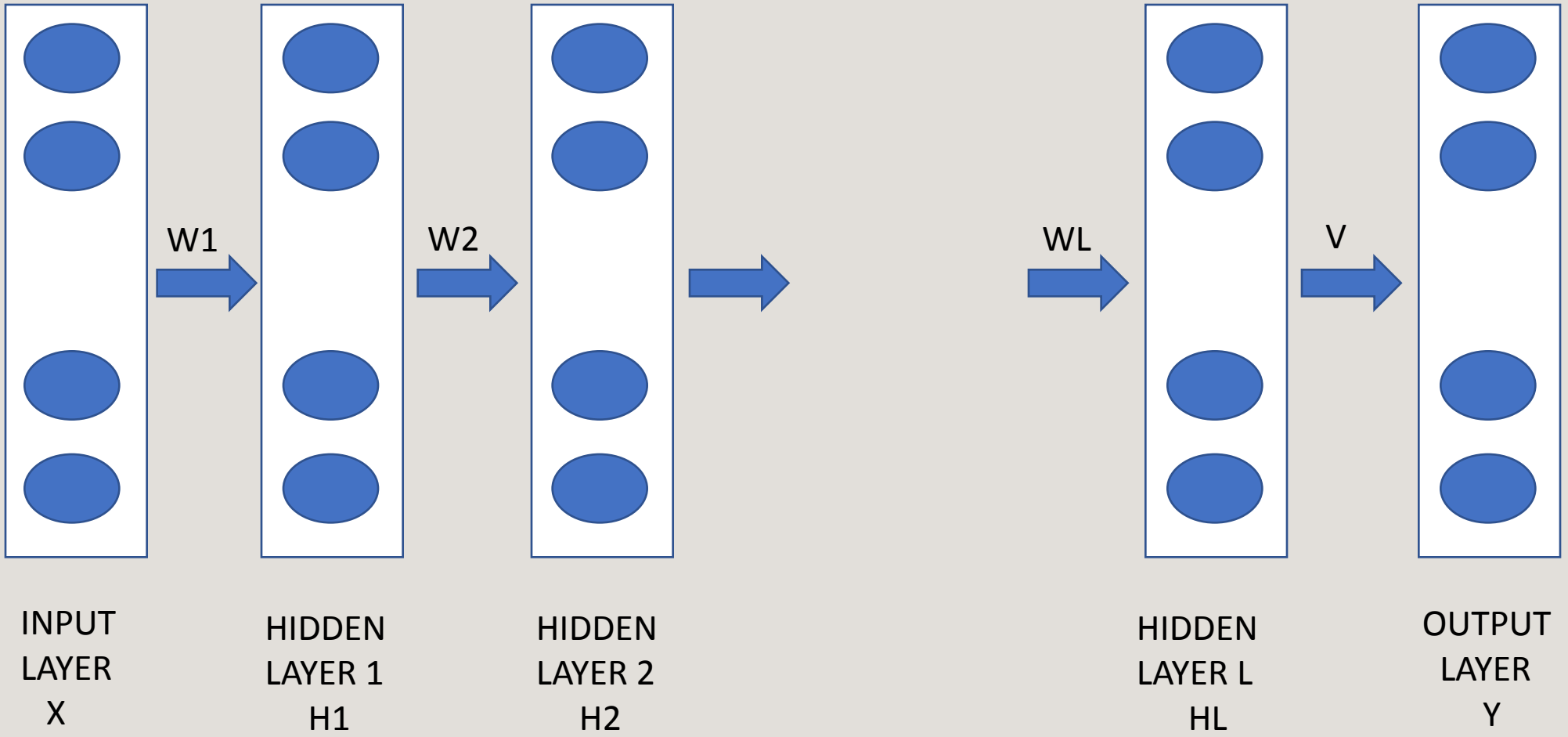


Biological Neuron versus Artificial Neural Network

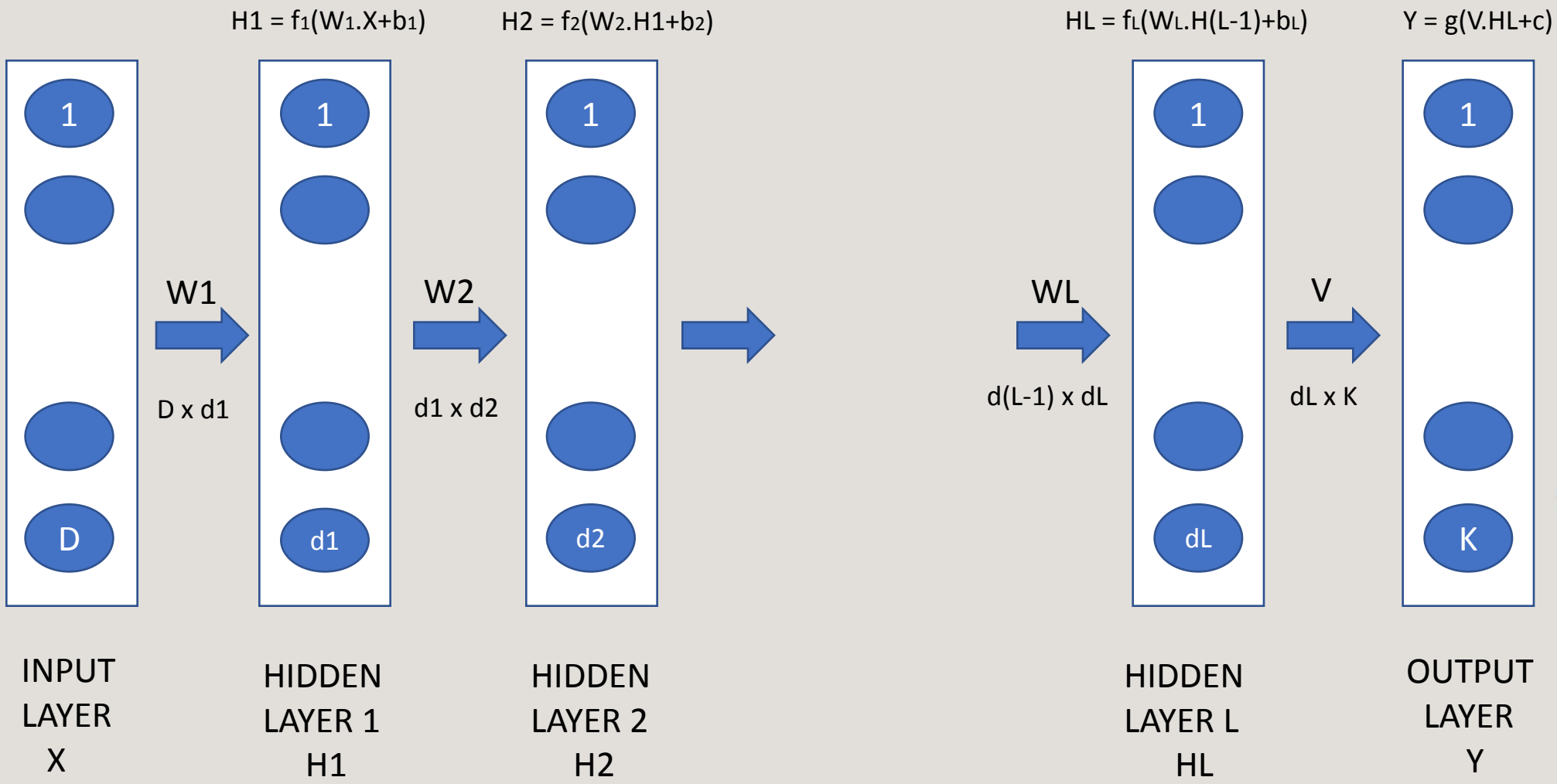


Source: Devopedia.com

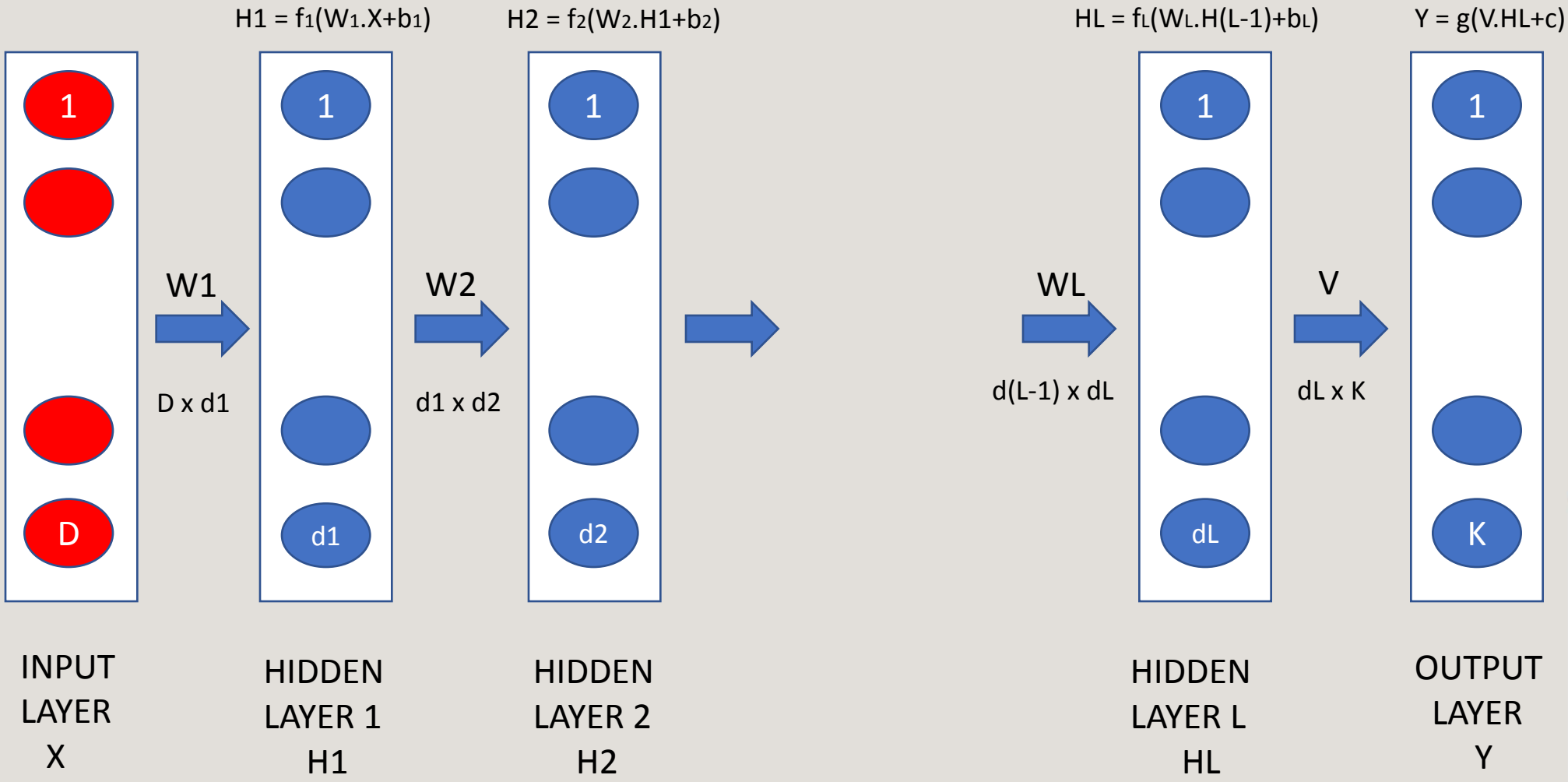
Multi-layer Neural Network



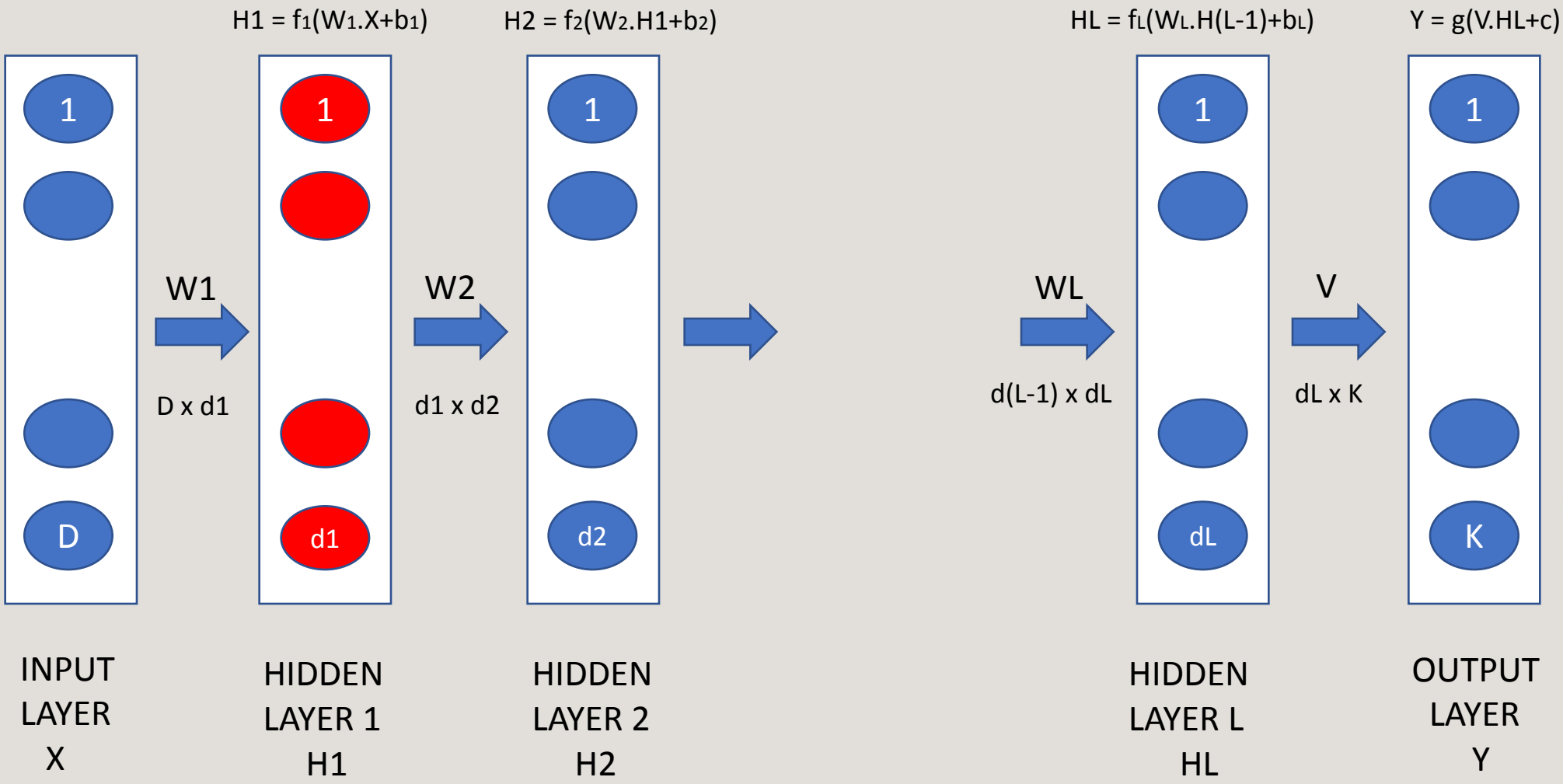
Multi-layer Neural Network



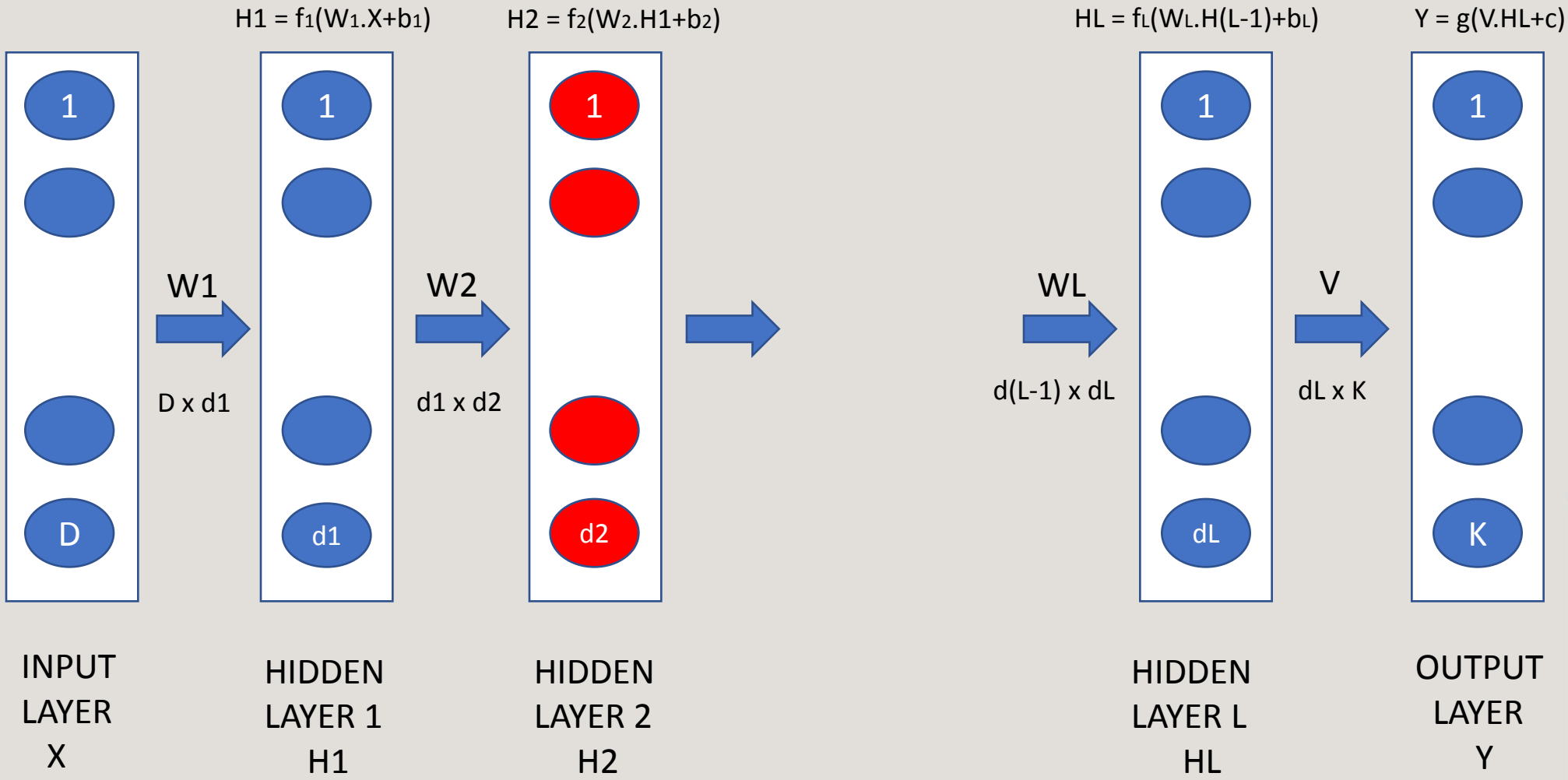
Multi-layer Neural Network



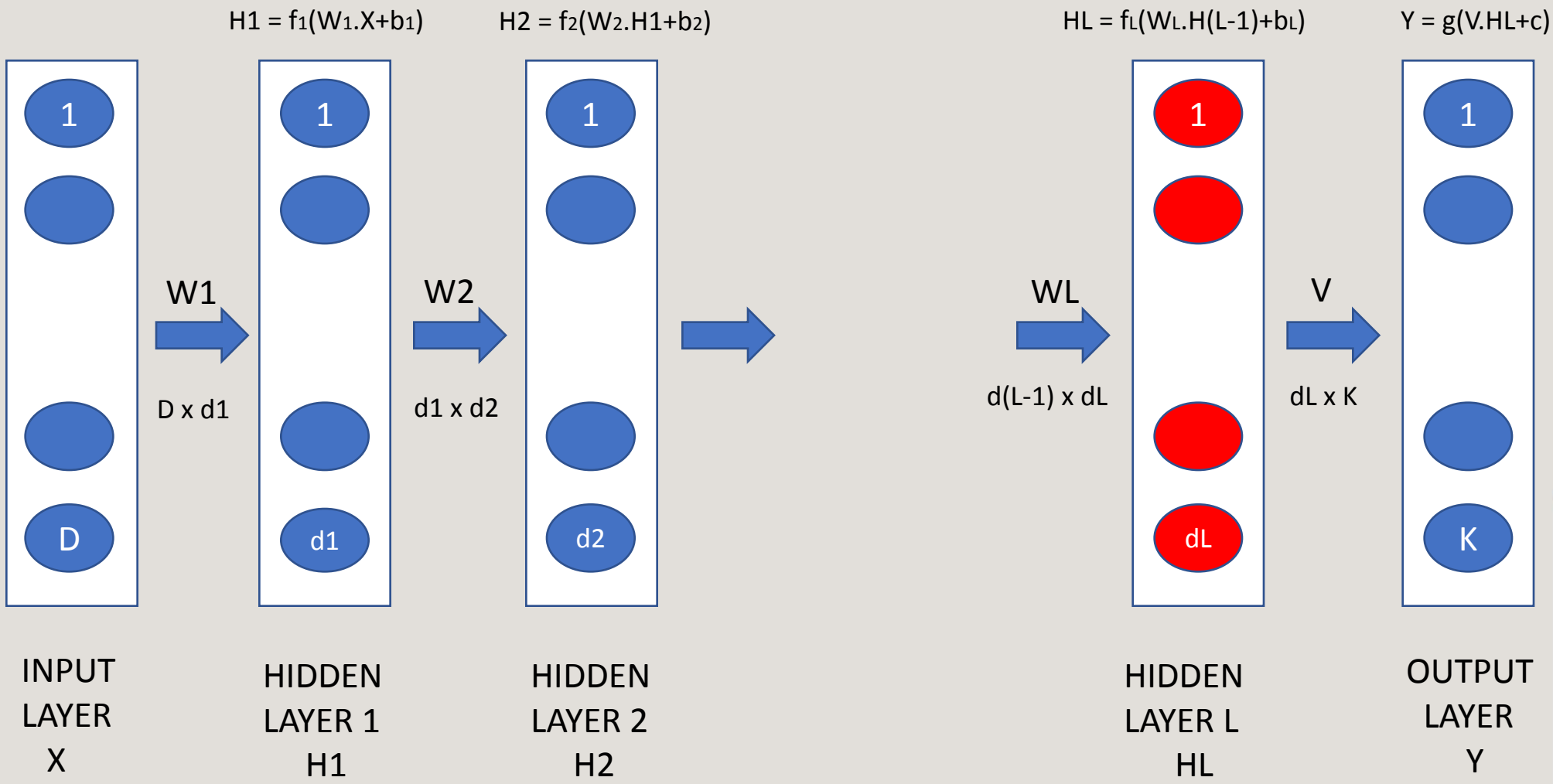
Multi-layer Neural Network



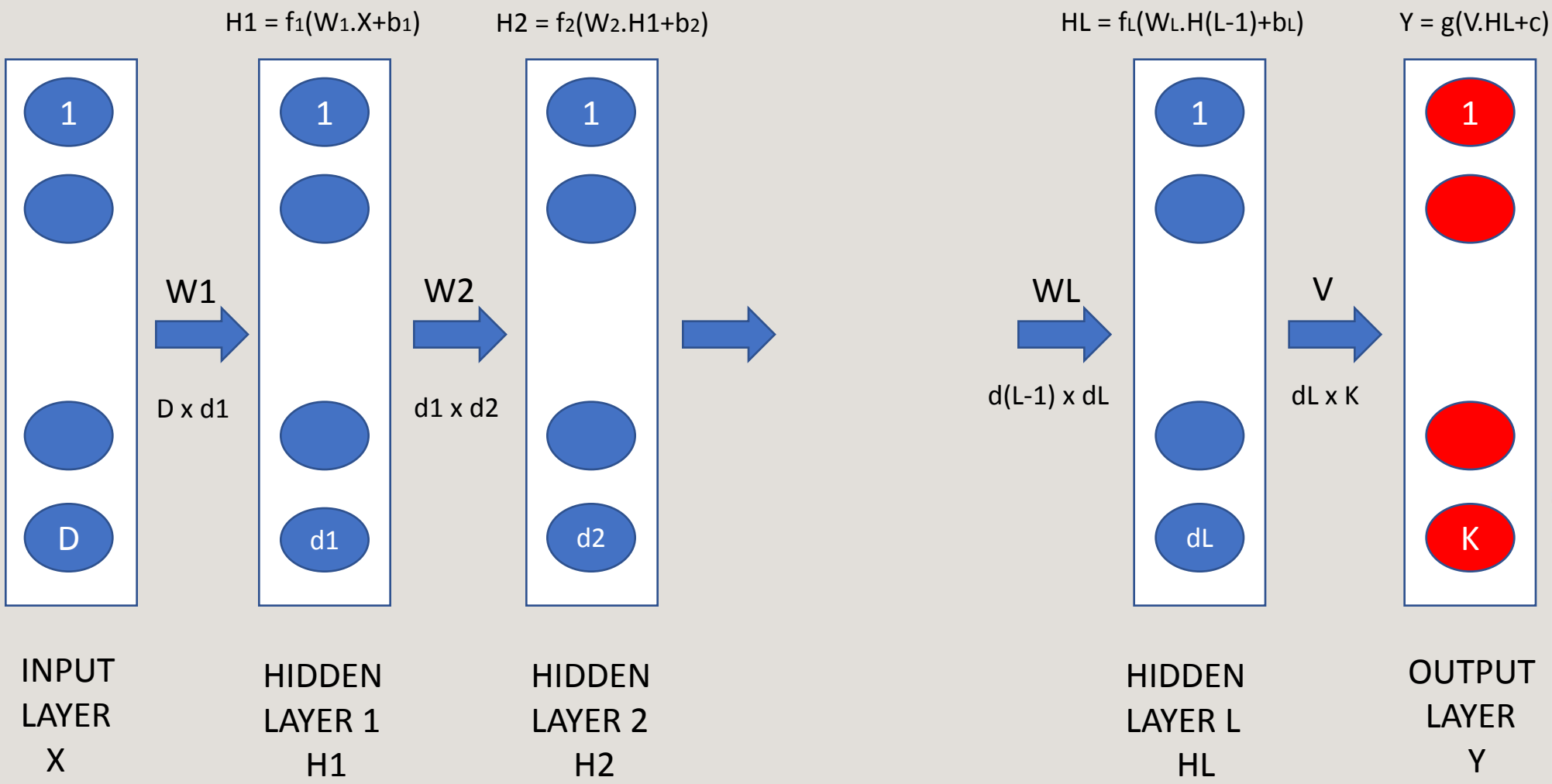
Multi-layer Neural Network



Multi-layer Neural Network

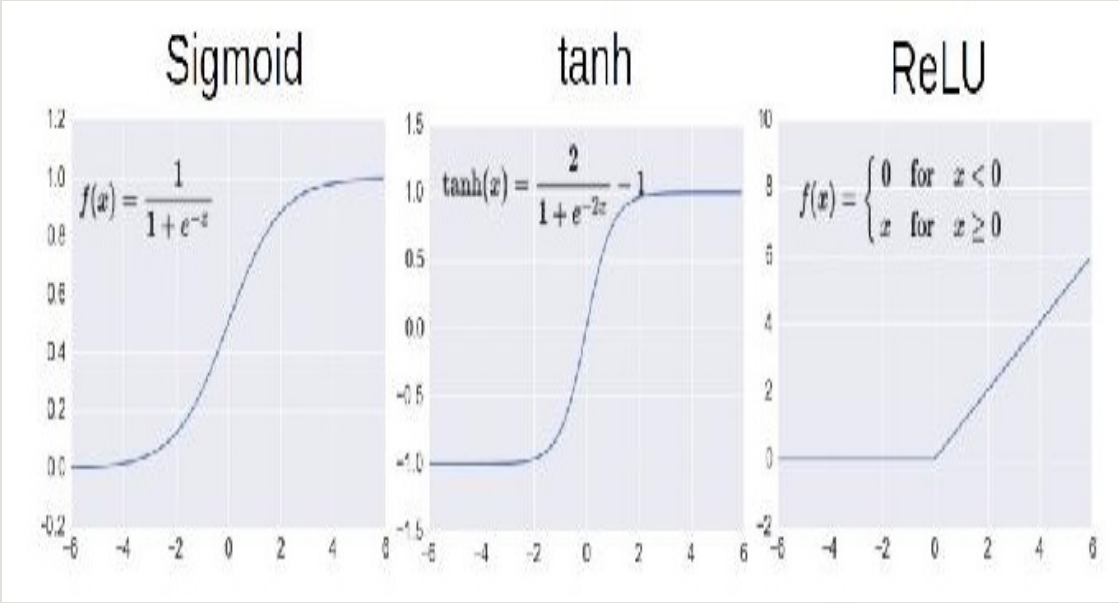


Multi-layer Neural Network



Design of Neural Network

- Neural networks have the following parameters
  1) Number of hidden layers
  2) Number of units in each hidden layer
  3) Types of activation functions in the hidden layers
- Hidden layers represent complex functions of input vector
- Functions are non-linear due to the activations
- Most common activations : Sigmoid, tanh and RELU
- These are chosen by the network designer



- However, the weight parameters W1, W2 etc have to be estimated from the data
- Consider loss function L(y, f(x)) where ‘f’ is the function represented by the neural network
- Choose the weight parameters to minimize the loss function

Estimation of Neural Networks Weights

Gradient descent used to adjust the weights
Required: derivative of loss function w.r.t. each weight!

Consider a simple neural network $Y = W.X$
Squared error $L(Y, t) = (Y-t)^2 = (W.X - t)^2 = (\sum_i W_i X_i - t)^2$

Derivative $\Delta L(w_j) = 2X_j(\sum_i W_i X_i - t)$
$W_j = W_j - \alpha \Delta L(W_j)$

$L(Y,t) = (Y-t)^2 = (f(W.X) - t)^2 = (f(\sum_i W_i X_i) - t)^2$

Derivative $\Delta L(w_j) = 2X_j f'(\sum_i W_i X_i)\ (f(\sum_i W_i X_i) - t)$

- For deep networks, the hidden weights too need to be updated!
- Weights are updated turnwise, from output layer to input layer

Backpropagation in Neural Networks

Gradient descent used to adjust the weights
Required: derivative of loss function w.r.t. each weight!

Consider a simple neural network $Y = W.X$
Squared error $L(Y, t) = (Y-t)^2 = (W.X - t)^2 = (\sum_i W_i X_i - t)^2$

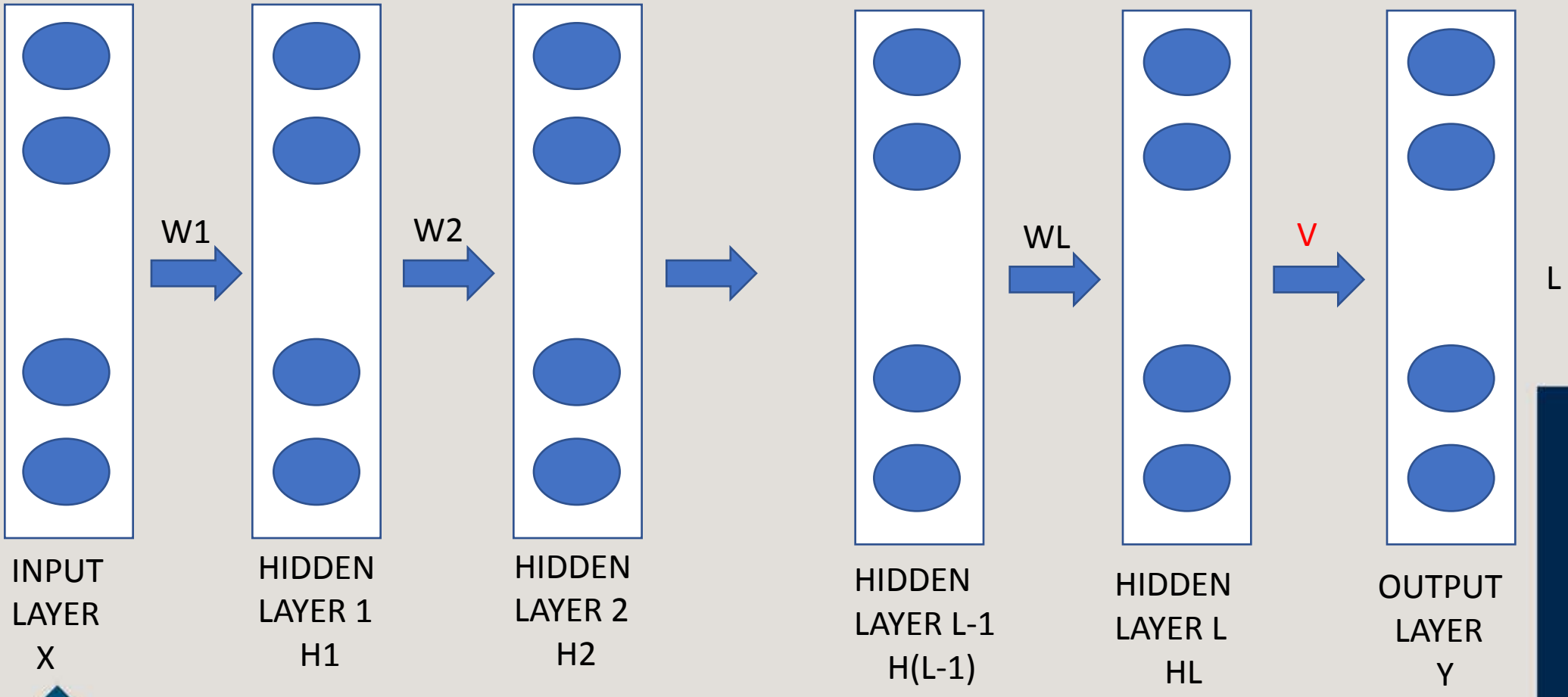
Derivative $\Delta L(w_j) = 2X_j(\sum_i W_i X_i - t)$
$W_j = W_j - \alpha \Delta L(W_j)$

$L(Y,t) = (Y-t)^2 = (f(W.X) - t)^2 = (f(\sum_i W_i X_i) - t)^2$

Derivative $\Delta L(w_j) = 2X_j f'(\sum_i W_i X_i)\ (f(\sum_i W_i X_i) - t)$

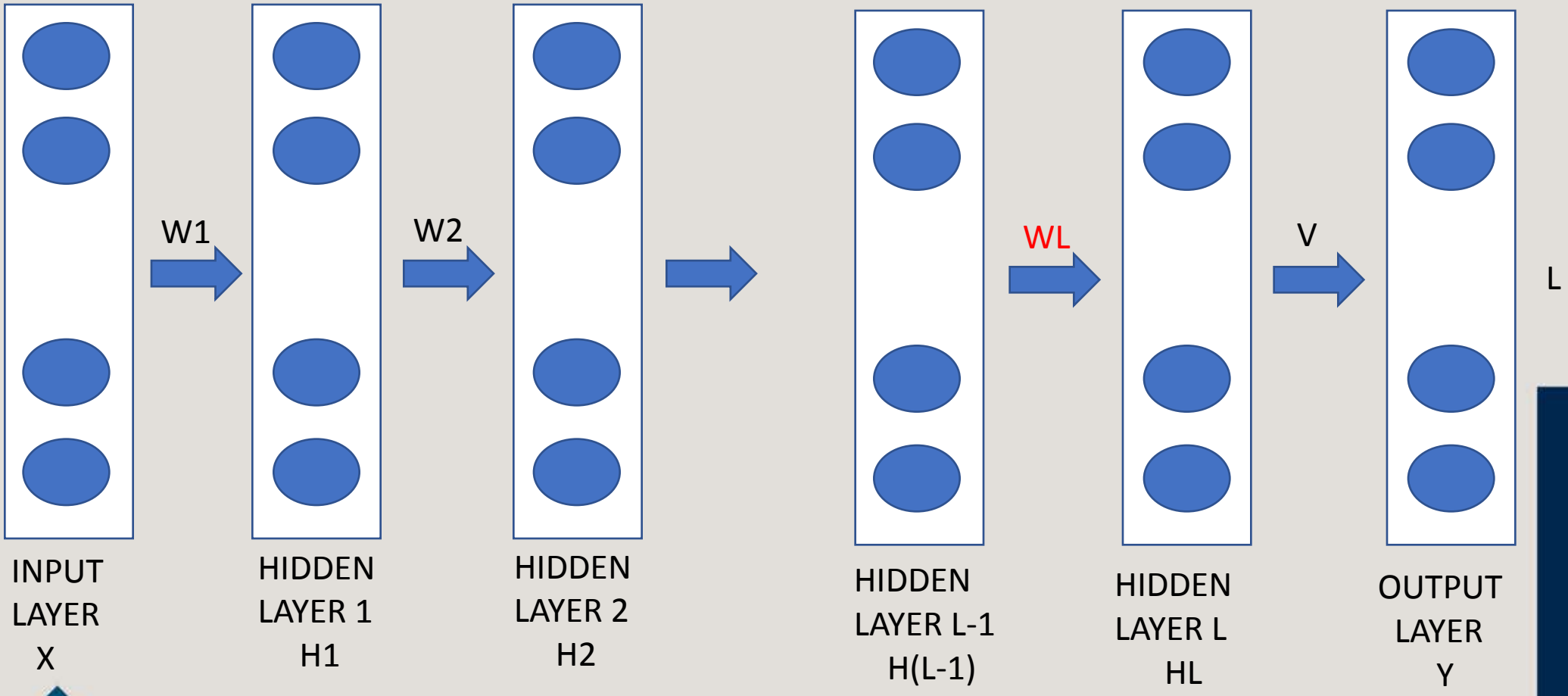
- For deep networks, the weights between intermediate layers too need to be updated!
- Weights are updated turnwise, from output layer to input layer

Backpropagation in Neural Networks

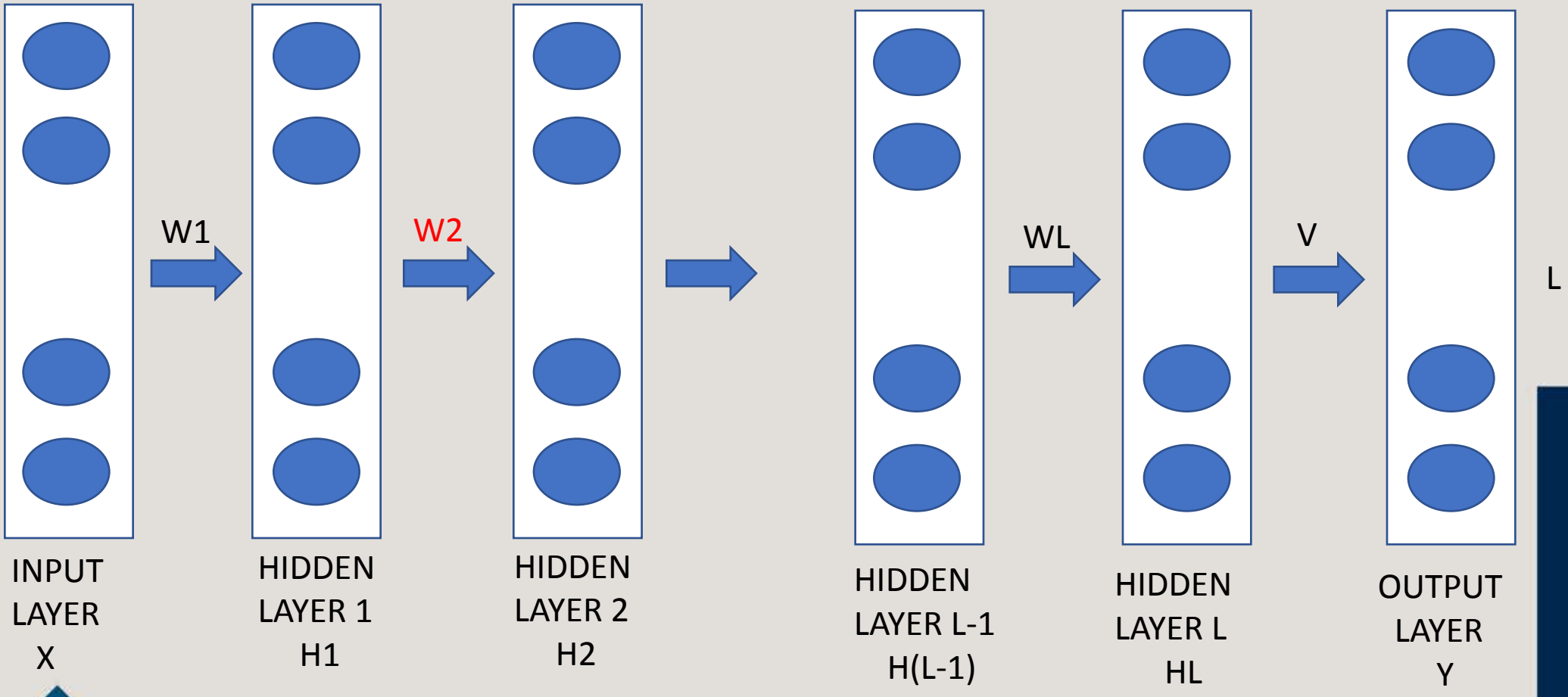


Backpropagation in Neural Networks

Use chain rule of differentiation and re-use the values already calculated



Backpropagation in Neural Networks



CONCLUSION

- A linear classifier can be expressed as a neural network with nodes and edges
- Each input feature expressed as input node, output as output node
- Addition of hidden layers allows us to express more complex decision boundaries
- Non-linear activation functions enable us to express arbitrarily complex functions
- In Feedforward mode computations carried out in steps from input to output
- Backpropagation: estimation of parameters by repeated use of chain rule of differentiation


THANK YOU

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ARTIFICIAL INTELLIGENCE FOR ECONOMICS

Lecture – 15

Topic: Deep Learning for Time Series Forecasting



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CONCEPTS COVERED

- **Neural Feature Extraction**
- **Recurrent Neural Networks**
- **Time Series Forecasting**
- **Challenges of Neural Networks and potential solutions**

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Neural Feature Extraction

- It is often challenging to find suitable feature representation for complex observations like images and time-series

- Traditional approach: apply various mathematical “filters” that may highlight interesting aspects
- Example: edges, corner points, textures in images, peaks/troughs in time series etc

- Finding suitable filters very tedious and non-intuitive

- Neural Networks may be used to extract a collection of “neural feature maps”

- Neural features: intermediate layer values in a deep neural network

- Obtained by standard neural network operations, can “simulate” mathematical filters if activation functions and weights chosen appropriately!

Convolution by Neural Networks

- Neural Network can simulate “convolution” operation – important in signal processing!
- Especially suitable for “structured” inputs such as images
- Helps to capture local properties of the input – represented by a “filter”

0	0	0	0
0	1	0	1
0	0	2	2
0	0	2	-1

*

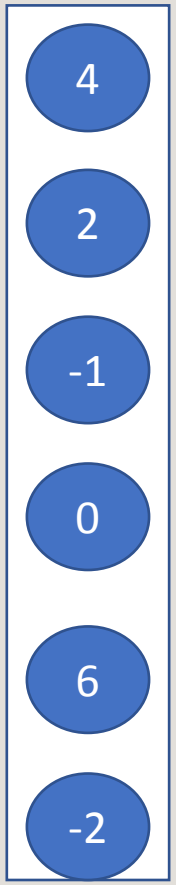
1	0	0
0	2	1
0	1	-1

=

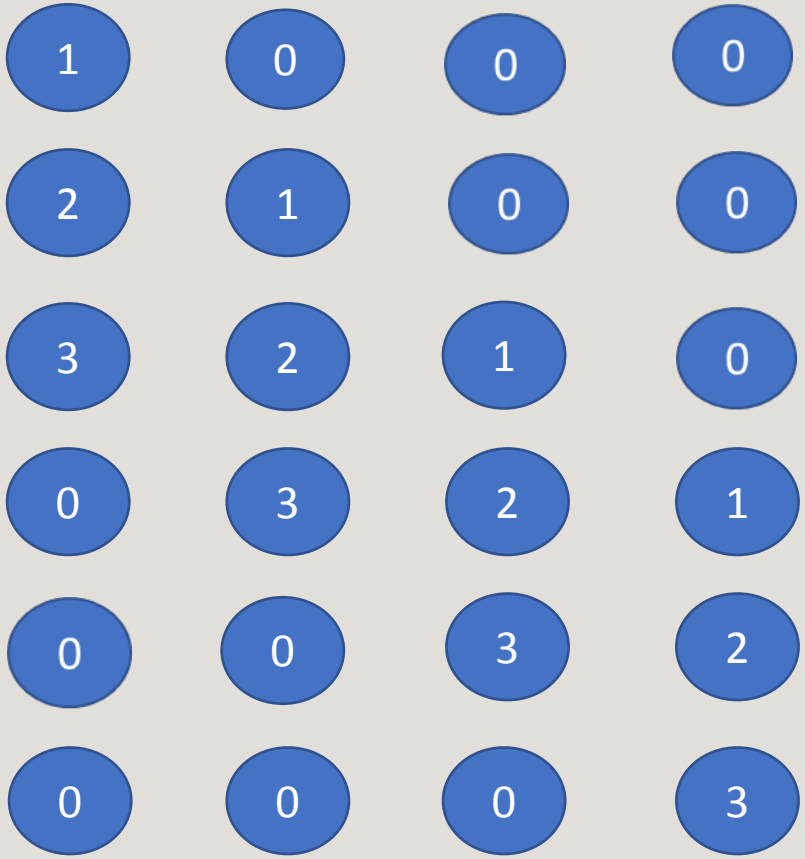
0	0	0	0
0	0	?	0
0	?	10	0
0	0	0	0

- Sparse connections between adjacent layers (except the last layer)
- Many edges between two layers have “shared weights” - reduces number of parameters
- The weight vector has repeating structures and many empty values
- Outputs can be useful features for representation!

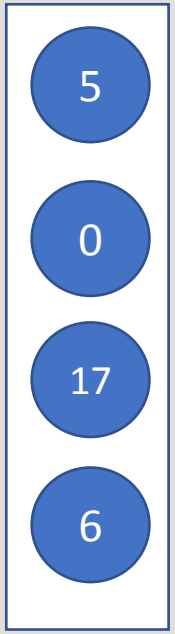
Convolution Operation with Neural Networks



INPUT



WEIGHT MATRIX

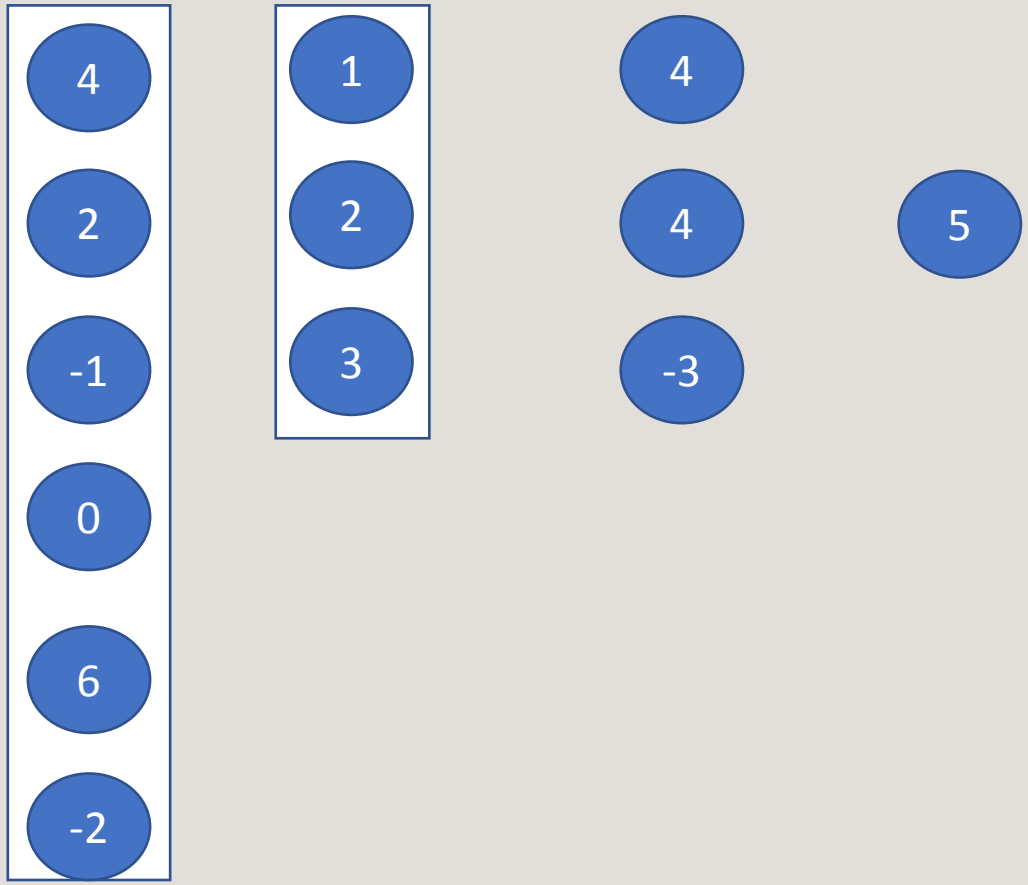


OUTPUT

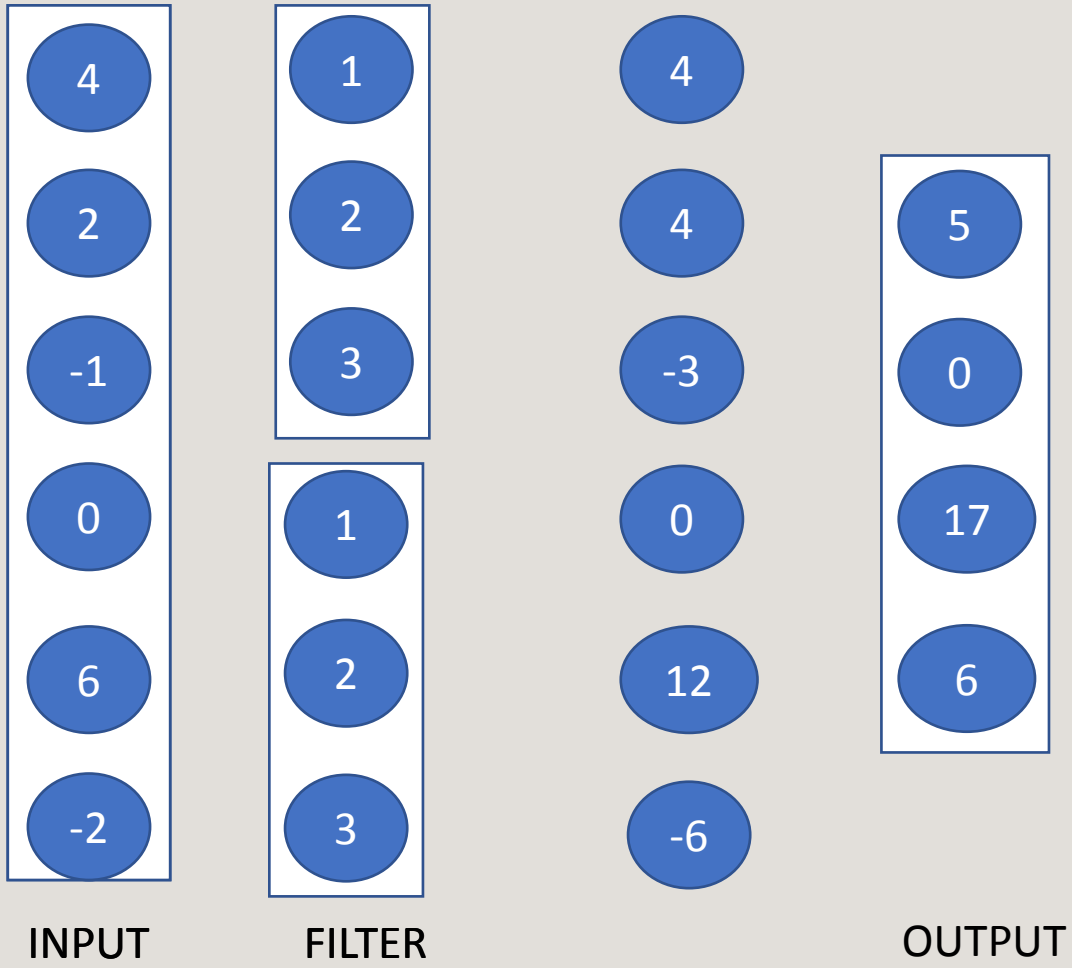


REPEATING STRUCTURE

Convolution Operation with Neural Network



Convolution Operation with Neural Network



Pooling Operation with Neural Network

- Blockwise operation on a vector or matrix
- Operation: max (most common), mean, sum
- Block size: user input

1	1	2	4
5	6	7	8
3	2	1	0
1	2	3	4

Block size:2
Stride: 2

6	8
3	4

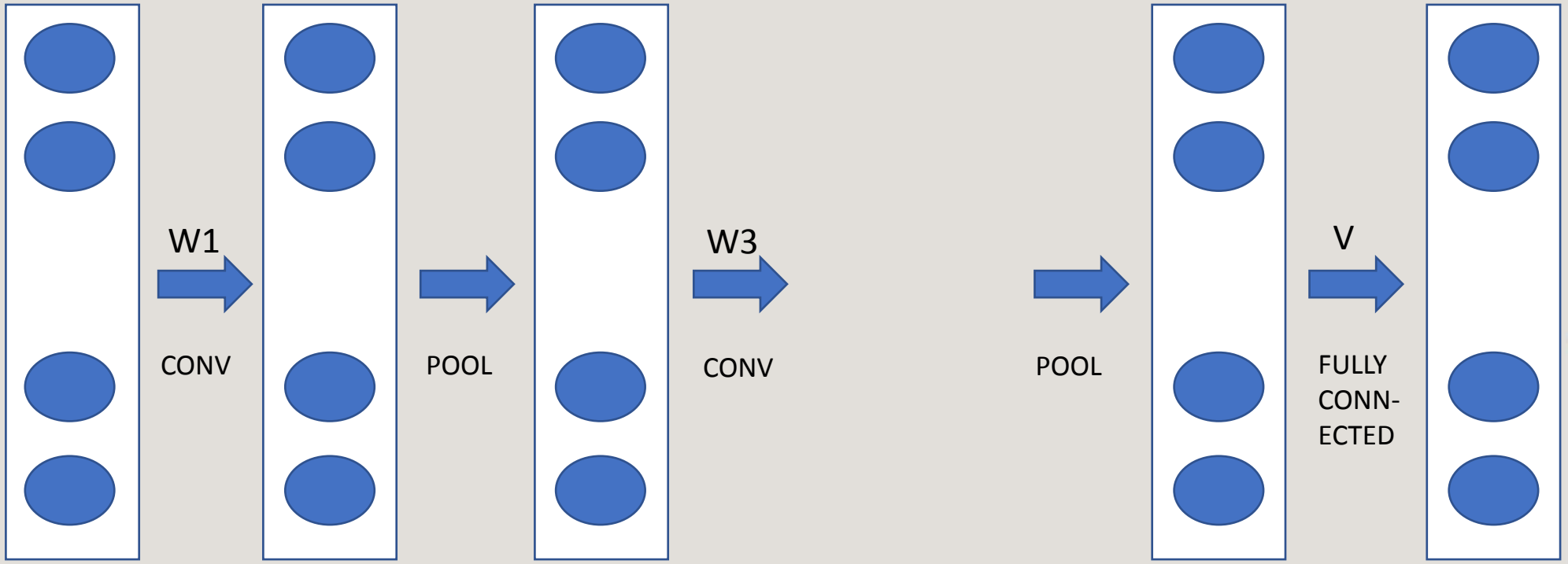
1	1	2	4
5	6	7	8
3	2	1	0
1	2	3	4

Block size:3
Stride: 1

7	8
7	8

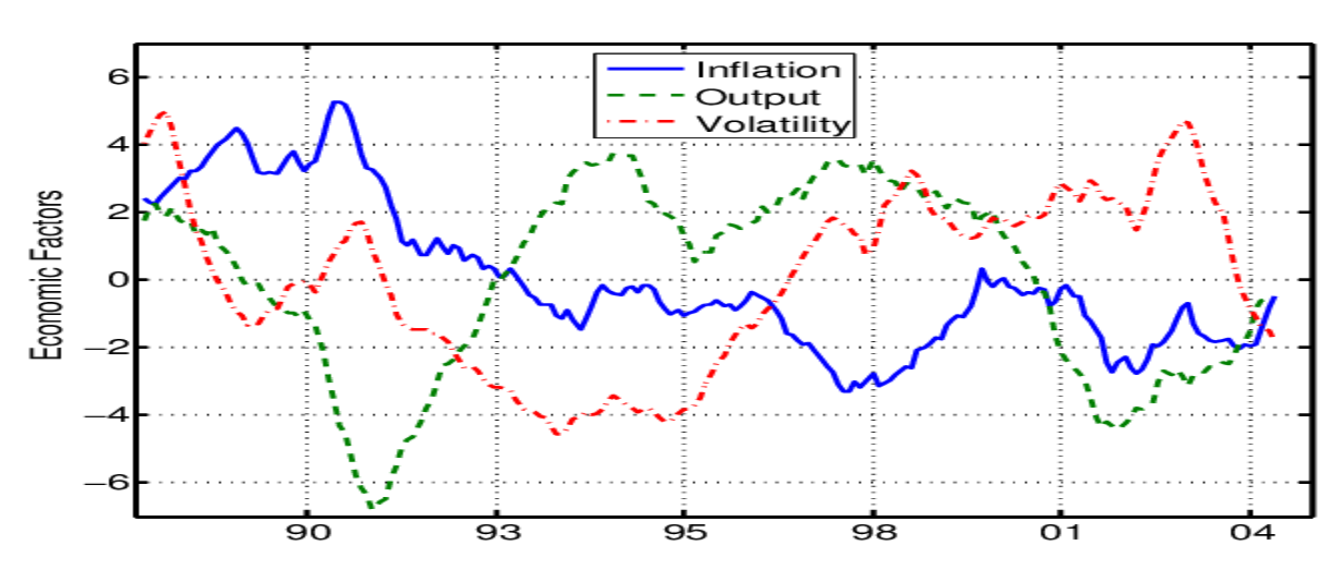
Pooling Operation with Neural Network

- A deep convolutional neural network has many “convolution layers”
- Usually, each convolutional layer is followed by a pooling layer
- In large neural networks, each convolution layer involves multiple convolution operations with multiple filters on the same input - creates “feature maps”



Time Series Data

- Sequence of observations, each observation marked with a time-stamp
- Observations can be scalar or vector-valued
- Frequently arises in Economics – stock values of a company, GDP of a country over time



- Time-series has 3 components: slow-moving (trend), fast-moving (periodic), random
- Time-series may have local peaks and troughs
- Aim: to extract “local patterns” for representation

Time Series Operations

- Short-term Forecasting: The dynamics of the model allows us to predict output using past inputs
- Long-term Forecasting: The next step's input itself can be predictable
- Time-series classification:
    1) Provide the entire time-series as input
    2) Hidden state forms an intermediate representation (vector/matrix/tensor)
    3) Classify based on this representation using another neural network
- Time-series translation/mapping: use the hidden state representation to generate a new time-series!

Time Series Forecasting

Given past values of a time-series X(1), X(2) …… X(t), predict the next value X(t+1)

Time-series typically shows “autocorrelation”, i.e. current values have correlation with past values

This property allows for forecasting future values in time-series

Simplest approach for forecasting: Autoregression

$X(t+1) = a1*X(t) + a2*X(t-1) + \ldots\ldots + aK*X(t-K+1)$ [K: window length]

Parameters a1, a2 …. Can be estimated by minimizing loss function (like linear regression)

Next step: latent variables - Temporal dynamics separated from observed values

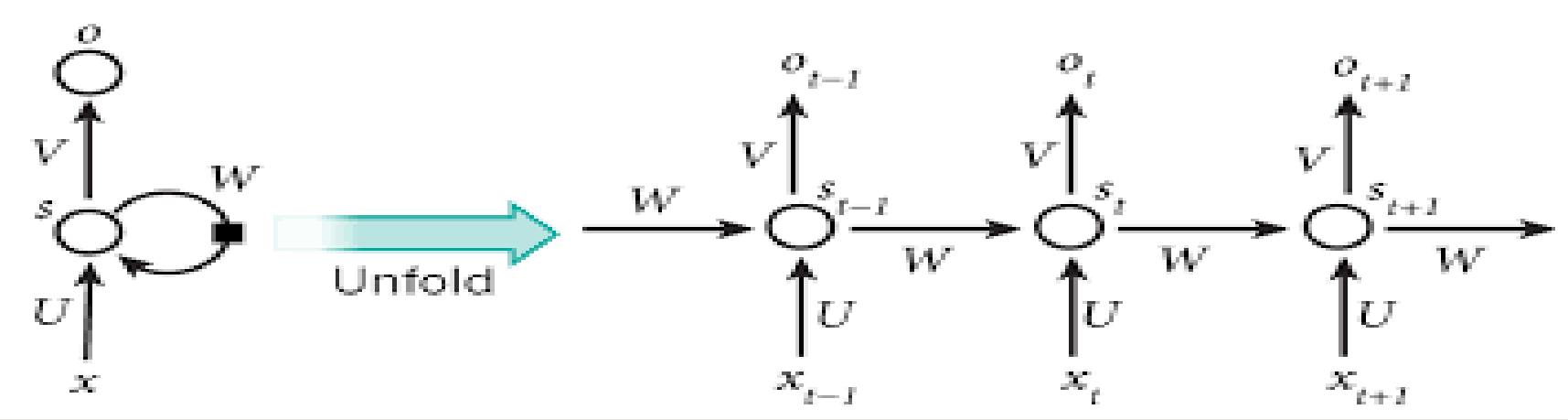
$S(t+1) = f(S(t), S(t-1), \ldots\ldots S(1))$

$X(t) = g(S(t))$

Need to estimate Z using past X to predict next values of X

Recurrent Neural Networks

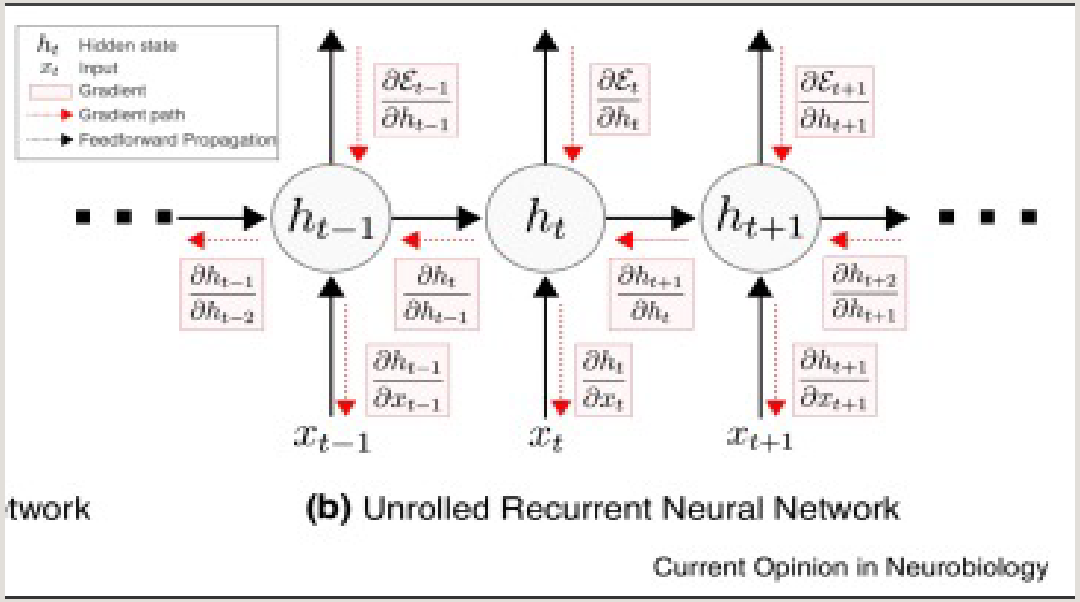
- Another hidden-state approach, hidden state value continuous
- Transition function can be complex, can use number of hidden state variables with instantaneous output
- Hidden state dynamics: S(t) = f(W*S(t-1) + U*X(t)) where f is nonlinear
- Output: O(t) = g(V*S(t)) where g is nonlinear



Lecun et al, Nature (2015)

Recurrent Neural Networks Training

- Training of RNN = estimation of parameters W and U
- Trained RNN can be used to forecast future values of time-series
- Parameters estimated using numerical methods like “Gradient Descent” – Backpropagation through time!



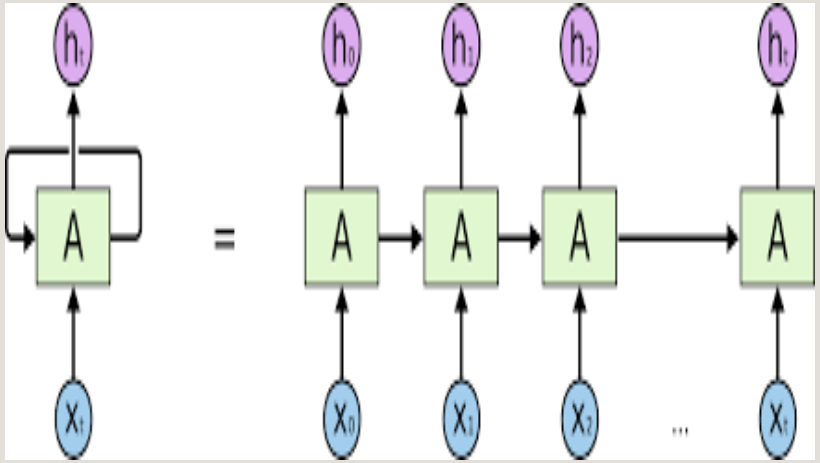
twork **(b)** Unrolled Recurrent Neural Network

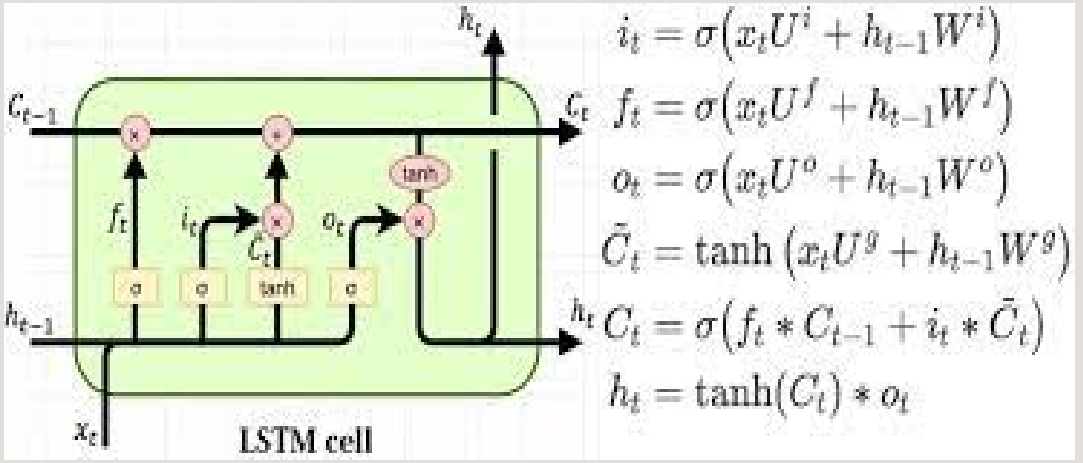
Current Opinion in Neurobiology

- Estimation of parameters difficult if training sequences are long!
- “Vanishing gradient” and “Exploding gradient” problems appear

Long Short Term Memory

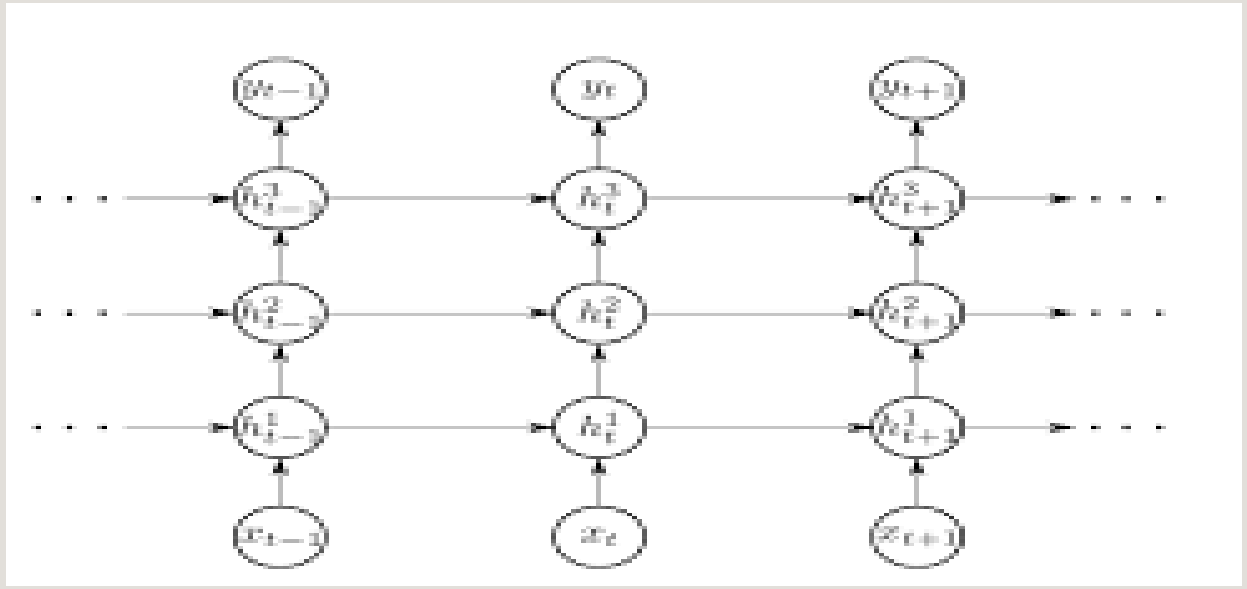
- The “hidden state” is more complex than RNN, can retain values over long sequences
- Cell state variable C(t): long-term memory, erased only if there is “forget” signal
- Short-term memory h(t): updated at each step

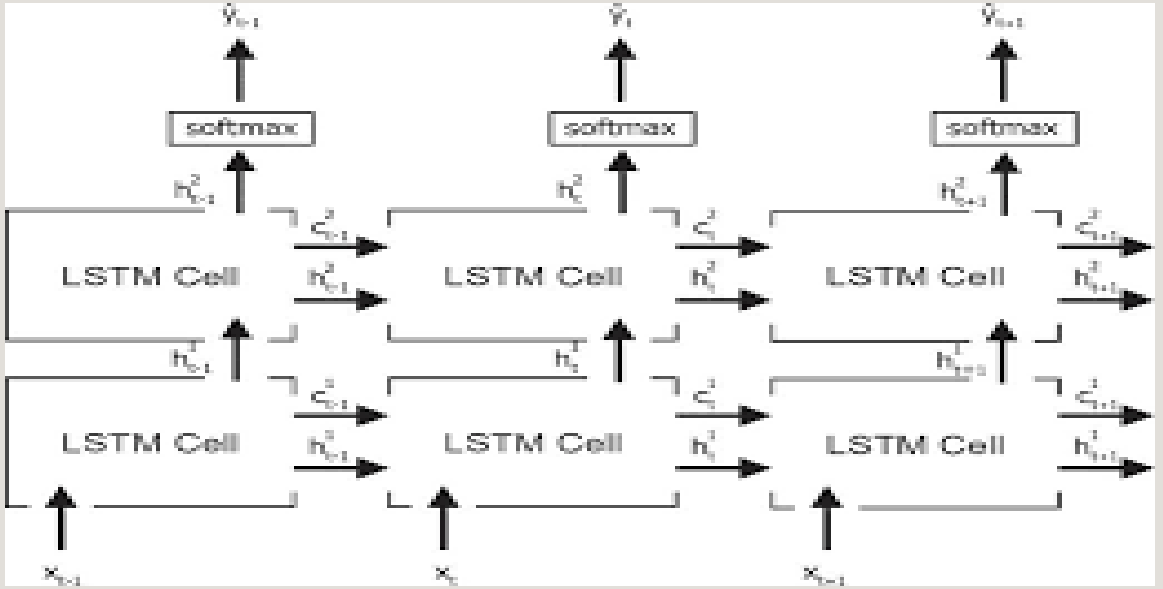




V. Savvas (2018)

Stacked Models





Richer representation possible with multiple hidden layers

Applications in Economics

- Forecasting: Given past values, how to predict the future values?
  - GDP/stock price forecasting
- Classification: Can we identify whether any given time-series belongs to a particular class?
  - Credit ratings to agencies/firms/economies
- Clustering: Can we identify which time-series are “similar” and group them together?
  - Groups of firms/countries having similar GDP behavior
- Segmentation: Can we identify “change-points” in a time-series?
  - Identify events that impact GDP/stock prices
- Alignment: Given two time-series that are “similar” but mis-aligned, can we align them?

CONCLUSION

- Neural Networks can be used to create feature representations of complex data
- Convolutional Neural Networks simulate the “convolution” operation
- Future values of a time-series can be forecasted based on past values
- Recurrent Neural Networks useful for time-series modelling and forecasting
- Suffer from computational problems, may “forget” earlier values in time-series
- Long-Short Term Memory: Solves many of these issues


THANK YOU